Mathematical Foundations of Computational Electrophysiology

A theorem-driven companion to neuronal

neuronal study notes

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How to read this book

We begin with ordinary sentences. The claim *Every spike belongs to a recorded neuron* can be true, false, or ambiguous if the meaning of *belongs* is not specified. Mathematics begins by removing ambiguity: we give names to objects, describe relations among them, and state exactly when a claim follows from its assumptions.

A *set* is a collection of objects. A *relation* describes which ordered pairs satisfy a condition. A *function* assigns one output to each admissible input. A signal is a function of time; a spike train can be represented by a counting measure; a covariance matrix represents pairwise relations among random variables; a neural trajectory is a function whose values lie in a vector space.

How to use the numbered results. Results 1–92 form the original indexed catalogue of named theorems, lemmas, inequalities, criteria and identities. The revised curriculum interleaves rigorous chapters on measures, Gaussian conditioning, electrode biophysics, digital filters, GLMs, renewal spectra, population identifiability, causal inference and neural dynamics. Classical sampling and random-matrix theorems are placed with their spectral prerequisites. The word *result* includes some identities and principles rather than only formal theorems. Compact catalogue entries do not replace proofs.

Notation. We write $x \in A$ for membership, $A \subseteq B$ for inclusion, $f : A \to B$ for a function, $\mathbb{E}[X]$ for expectation, $\mathrm{Cov}(X, Y)$ for covariance, and X^* for conjugate transpose.

An example before advanced mathematics. Let $U = \{u_1, u_2, u_3\}$ be the set of identified units and let $T = [0, 10)$ seconds. A spike event is an ordered pair $(u, t) \in U \times T$. The relation $R \subseteq U \times T$ records which spikes occurred. A count function maps each unit $u \in U$ to its number of spike times in T. This is the progression from a sentence to sets, relations, functions, numbers, and finally a statistical model.

Scientific caution. The repository's `matlab/README.md` explicitly describes its MATLAB methods as exploratory and not validated for scientific inference. The mathematical validity of a formula is separate from the adequacy of the assumptions required to infer physiology from data.

Reading order after the review. The physical forward problem—membrane current to extracellular potential to digitized signal—now precedes its spectral and inferential models. For a chapter-by-chapter accounting of resolved and outstanding gaps, see the revision matrix linked in the LaTeX README. A complete proof of every theorem is not claimed.

Part I

Language and mathematical foundations

Chapter 1

From ordinary statements to mathematical structures

A mathematical proof is a sequence of statements whose conclusion follows from explicitly declared assumptions. Start with the sentence *If a spike occurred in an interval, the interval contains a spike.* The implication is meaningful only after the interval and the membership relation have been defined.

Definition 1.1. A *relation* from A to B is a subset $R \subseteq A \times B$. It is a *function* $f : A \to B$ when every $a \in A$ occurs as the first coordinate of exactly one pair $(a, b) \in R$.

For example, a spike train of a unit is a subset of its observation time interval. A classification of spike waveforms by cluster labels is a function only when every input waveform receives exactly one label. This does not imply that the label identifies a physiological neuron.

Result 1: De Morgan's laws. *Precise setting:* For propositions P, Q, $\neg(P \wedge Q) \Leftrightarrow (\neg P) \vee (\neg Q)$; for sets, $(A \cup B)^c = A^c \cap B^c$, relative to an ambient set.

Why it matters here: Transform selection criteria and complements of detected event sets.

Result 2: Quantifier duality. *Precise setting:* For any predicate P, $\neg(\forall x\, P(x)) \Leftrightarrow \exists x\, \neg P(x)$; likewise $\neg(\exists x\, P(x)) \Leftrightarrow \forall x\, \neg P(x)$.

Why it matters here: Express universal reliability claims and their counterexamples.

Result 3: Mathematical induction (principle). *Precise setting:* If $P(0)$ holds and $P(n) \Rightarrow P(n+1)$ for $n \in \mathbb{N}$, then $P(n)$ holds for all $n \in \mathbb{N}$.

Why it matters here: Prove recurrences, finite sums, binning bounds, and iterative invariants.

Result 4: Recursion theorem on the natural numbers. *Precise setting:* With a starting value and a step map, there is a unique sequence satisfying the corresponding recursive definition in the standard set-theoretic setting.

Why it matters here: Construct discrete filters, numerical solvers, and event counters.

Result 5: Equivalence relations and partitions. *Precise setting:* Reflexive, symmetric, transitive relations correspond to partitions into disjoint nonempty equivalence classes.

Why it matters here: Distinguish clustering as a partition from physiological identification.

Result 6: Well-ordering principle. *Precise setting:* Every nonempty subset of $\mathbb{N}$ has a least element.

Why it matters here: Prove termination of monotone integer-index algorithms.

Result 7: Cantor–Schroeder–Bernstein theorem. *Precise setting:* If injections $A \to B$ and $B \to A$ exist, then a bijection $A \to B$ exists.

Why it matters here: Understand cardinality and information-preserving relabelings.

Result 8: Cantor's diagonal argument. *Precise setting:* There is no surjection $\mathbb{N} \to (0, 1)$; the real interval is uncountable.

Why it matters here: Understand the distinction between finite sampled arrays and continuum models.

Result 9: Least-upper-bound property (completeness axiom for $\mathbb{R}$). *Precise setting:* Every nonempty subset of $\mathbb{R}$ bounded above has a real supremum.

Why it matters here: Build limits, continuity, completeness, and convergence of numerical approximations.

Result 10: Archimedean property and density of $\mathbb{Q}$. *Precise setting:* For real x there is an integer $n > x$; between any $a < b$ in $\mathbb{R}$ lies some $q \in \mathbb{Q}$.

Why it matters here: Approximate continuous times and amplitudes by rational values, without confusing approximation with exact sampling.

Example 1.2 (Equivalence classes of spike timestamps)**.** For a fixed bin width $\Delta > 0$, let $s \sim t$ when $\lfloor s/\Delta \rfloor = \lfloor t/\Delta \rfloor$. This relation is reflexive, symmetric, and transitive. Each equivalence class represents one temporal bin, not one spike. A spike-count sequence counts members of each class.

Interpretation warning 1.3. Membership ($x \in A$), inclusion ($A \subseteq B$), and association under a binary relation (xRy) are distinct concepts. An observed cross-correlation is not evidence that a causal or anatomical edge exists.

1.1 Proofs of selected foundational results

The catalogue entries above state results compactly. This section proves those whose arguments are short enough to be reproduced by hand and whose structure reappears later in the book.

Proof of Result 1 for sets. Let U be the ambient set and $x \in U$. Then $x \in (A \cup B)^c$ iff $\neg(x \in A \vee x \in B)$ iff $(x \notin A) \wedge (x \notin B)$ by the propositional De Morgan law, iff $x \in A^c \cap B^c$. Since the two sets have the same members they are equal. The propositional law itself is verified by its four-row truth table. □

Proposition 1.4 (Result 6 implies Result 3)**.** *The well-ordering principle implies the principle of induction.*

Proof. Assume $P(0)$ and $P(n) \Rightarrow P(n+1)$ for all n, and suppose the set $F = \{n \in \mathbb{N} : \neg P(n)\}$ is nonempty. Let m be its least element. Then $m \neq 0$ because $P(0)$ holds, so $m - 1 \in \mathbb{N}$ and $m - 1 \notin F$, i.e. $P(m - 1)$ holds. The induction step gives $P(m)$, contradicting $m \in F$. □

Proof of Result 8. Suppose $g : \mathbb{N} \to (0, 1)$ is surjective and write each $g(n)$ by its decimal expansion $0.d_{n1}d_{n2}\ldots$, choosing the expansion that does not end in repeated 9's. Define $y = 0.e_1e_2\ldots$ with $e_n = 5$ if $d_{nn} \neq 5$ and $e_n = 4$ if $d_{nn} = 5$. Then $y \in (0, 1)$, its expansion is of the chosen kind, and it differs from $g(n)$ in the nth digit for every n. Hence y is not in the image of g. □

Proof sketch of Result 7. Let $f : A \to B$ and $g : B \to A$ be injective. Call $a \in A$ an *A-stopper* if, tracing preimages $a,\ g^{-1}(a),\ f^{-1}(g^{-1}(a)), \ldots$ as far as they exist, the chain terminates in $A \setminus g(B)$. Define $h(a) = f(a)$ for A-stoppers and $h(a) = g^{-1}(a)$ otherwise. One checks that every non-stopper lies in $g(B)$, that h is injective on each of the two pieces, and that the two images partition B. Thus h is a bijection. □

Result 7 is used implicitly whenever one argues that two index sets "have the same size" by exhibiting encodings in each direction, for example that finite spike rasters over a countable set of bins can be enumerated.

1.2 Relations as data structures

Most data objects in an electrophysiology pipeline are relations or functions whose properties should be checked rather than assumed.

- A spike table with columns (unit, time) is a relation $R \subseteq U \times \mathbb{R}$. It is the graph of a function $U \to \mathbb{R}$ only if each unit fires exactly once.
- A channel map is a function from channel indices to physical positions. It must be injective (no two channels at the same place) for current-source density estimates to make sense.
- A trial table is a function from trial indices to (condition, onset) pairs. If it is not injective in onset, two trials share a time window and their data are not independent.
- A cluster assignment is a function from detected waveforms to labels, equivalently a partition (Result 5). A *merge* of two clusters replaces the partition by a coarser one; a *split* replaces it by a finer one.

Proposition 1.5 (Agreement between two sortings)**.** *Let $\mathcal{P}$ and $\mathcal{Q}$ be two partitions of the same finite set of detected spikes S. The meet $\mathcal{P} \wedge \mathcal{Q} = \{P \cap Q : P \in \mathcal{P},\ Q \in \mathcal{Q},\ P \cap Q \neq \varnothing\}$ is a partition of S that refines both.*

Proof. Each $s \in S$ lies in exactly one P and one Q, hence in exactly one $P \cap Q$. Blocks are nonempty by construction and pairwise disjoint because distinct pairs (P, Q) differ in at least one coordinate, whose blocks are disjoint. Each $P \cap Q$ is contained in P and in Q. □

The confusion matrix $M_{PQ} = |P \cap Q|$ used to compare sorters is the table of block sizes of the meet. Agreement indices such as the adjusted Rand index are functions of this table and answer the question "how similar are these two partitions?", not "which partition corresponds to neurons?".

Example 1.6 (Bins as equivalence classes, continued)**.** For the binning relation $s \sim t \iff \lfloor s/\Delta \rfloor = \lfloor t/\Delta \rfloor$, the quotient map $t \mapsto \lfloor t/\Delta \rfloor$ is a function from times to integers. It is not injective, so binning cannot be undone. Two spikes of different units in the same bin

are *coincident at resolution* Δ; the same pair may be non-coincident at resolution $\Delta/2$. Coincidence is therefore a relation that depends on a parameter, and every synchrony claim must state that parameter.

Exercises for this chapter

Exercise 1.7. Prove the set form of De Morgan's law for an arbitrary indexed family $(A_i)_{i \in I}$.

Exercise 1.8. Show that the induction principle implies the well-ordering principle, completing the equivalence of Results 3 and 6.

Exercise 1.9. Prove that the set of finite subsets of $\mathbb{N}$ is countable, and that the set of all subsets of $\mathbb{N}$ is not.

Exercise 1.10. Two sorters produce partitions of 10,000 spikes into 12 and 15 clusters. What is the largest and the smallest possible number of blocks in their meet?

Chapter 2

The measure-theoretic vocabulary behind probability

A probability model is not merely a formula for averages. It states *which outcomes exist*, *which events can be assigned probabilities*, and *which calculations are allowed*. This chapter provides the missing hypotheses for Results 21–24, 40–46, and 71–76.

Definition 2.1 (Sigma-algebra and measure)**.** For a set Ω, a σ-algebra $\mathcal{F}$ is a family of subsets containing Ω, closed under complements and countable unions. A measure $\mu : \mathcal{F} \to [0, \infty]$ satisfies $\mu(\varnothing) = 0$ and countable additivity over disjoint events. A probability measure has $\mu(\Omega) = 1$. A measure is *σ-finite* if Ω is a countable union of measurable sets of finite measure.

Definition 2.2 (Random variable and expectation)**.** A real-valued random variable X is a measurable map $(\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$: for every Borel set B, its preimage $X^{-1}(B)$ is an event. If $\int |X| \, d\mathbb{P} < \infty$, define $\mathbb{E}X = \int X \, d\mathbb{P}$. The space L^2 consists of equivalence classes of variables with $\mathbb{E}|X|^2 < \infty$; equality almost surely is identified. Its inner product is $\langle X, Y \rangle = \mathbb{E}[X\overline{Y}]$. A *Hilbert space* is a complete inner-product space; L^2 is one. Completeness is what makes orthogonal projection possible.

Proposition 2.3 (Conditional expectation is a projection)**.** *For $X \in L^2$ and a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$, the conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is characterized by being $\mathcal{G}$-measurable and satisfying $\mathbb{E}[(X - \mathbb{E}[X \mid \mathcal{G}])Z] = 0$ for all bounded $\mathcal{G}$-measurable Z. It is the orthogonal projection of X onto the closed subspace $L^2(\mathcal{G})$.*

Proof. The defining conditional-expectation identity gives orthogonality for bounded Z. Truncation followed by L^2 approximation extends it to $Z \in L^2(\mathcal{G})$. The projection theorem then characterizes the minimizer of $\mathbb{E}[(X - Z)^2]$ among such Z. □

This explains why a conditional firing-rate prediction is an optimal mean-square prediction *relative to the variables conditioned upon*, not relative to every unobserved brain state.

Definition 2.4 (Stationarity, ergodicity, and predictability)**.** A process X_t is *wide-sense stationary* if its mean is constant and $\mathrm{Cov}(X_t, X_s)$ depends only on $t - s$. It is *strictly stationary* if all finite-dimensional joint laws are time-shift invariant. An invariant process is *ergodic* when invariant events have probability zero or one. A predictable intensity $\lambda(t \mid \mathcal{H}_{t-})$ is measurable relative to information available *just before* t; this forbids predicting a spike using its own occurrence. The time-integrated intensity is called its *compensator* when the Doob–Meyer conditions are satisfied. For a regular parametric model p_θ, the *score* is

$U_\theta = \nabla_\theta \log p_\theta(X)$ and *Fisher information* is $\mathcal{I}(\theta) = \mathbb{E}_\theta[U_\theta U_\theta^\top]$, under assumptions justifying its finiteness and calculus.

Proposition 2.5 (Law of total variance revisited)**.** *For $X \in L^2$,*

$$\operatorname{Var}(X) = \mathbb{E}[\operatorname{Var}(X \mid \mathcal{G})] + \operatorname{Var}(\mathbb{E}[X \mid \mathcal{G}]).$$

Proof. Write $X - \mathbb{E}X = (X - M) + (M - \mathbb{E}X)$ where $M = \mathbb{E}[X \mid \mathcal{G}]$. The cross term vanishes by orthogonality. Taking squared expectations yields the identity. □

Example 2.6 (LFP recording versus ensemble)**.** If a stationary and ergodic model describes a long LFP trace, its time average can estimate the ensemble mean under integrability. Stationarity alone does not justify equating those means, nor does increasing the sample rate create independent trials.

2.1 Modes of convergence and their counterexamples

For random variables on one probability space, four modes of convergence are in common use:

- *almost sure*: $\mathbb{P}(X_n \to X) = 1$;
- *in L^p*: $\mathbb{E}|X_n - X|^p \to 0$;
- *in probability*: $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$ for every $\varepsilon > 0$;
- *in distribution*: $\mathbb{P}(X_n \le x) \to \mathbb{P}(X \le x)$ at continuity points of the limit.

Almost sure convergence and L^p convergence each imply convergence in probability, which implies convergence in distribution. No other implication holds in general.

Proposition 2.7. *L^p convergence implies convergence in probability.*

Proof. Markov's inequality applied to $|X_n - X|^p$ gives $\mathbb{P}(|X_n - X| > \varepsilon) \le \mathbb{E}|X_n - X|^p / \varepsilon^p \to 0$. □

Example 2.8 (Convergence without convergence of means)**.** On $\Omega = (0, 1)$ with Lebesgue measure let $X_n = n\mathbf{1}_{(0,1/n)}$. Then $X_n \to 0$ almost surely, but $\mathbb{E}X_n = 1$ for all n. The spike-count analogue is a rare, large burst: it vanishes in probability but keeps a fixed contribution to the mean.

Example 2.9 (Convergence in probability without almost sure convergence)**.** Enumerate the dyadic intervals $[0, 1], [0, \frac{1}{2}], [\frac{1}{2}, 1], [0, \frac{1}{4}], \ldots$ and let X_n be the indicator of the nth one. The lengths tend to zero, so $X_n \to 0$ in probability, but every ω lies in infinitely many of the intervals, so $X_n(\omega)$ does not converge.

Consistency of an estimator is usually stated in probability; strong consistency is almost sure. A central limit theorem is a statement in distribution, and says nothing about any particular realization.

2.2 Product measures and independence

Independence of X and Y means that their joint law is the product of the marginals: $\mathbb{P}_{(X,Y)} = \mathbb{P}_X \otimes \mathbb{P}_Y$. For σ-finite measures the product measure is unique on the product σ-algebra, and the Tonelli and Fubini theorems (Chapter 11) license computing expectations of functions of independent variables by iterated integration. Independence of a whole family requires the product property for every finite subfamily.

2.3 Filtrations, stopping times, and martingales

A *filtration* $(\mathcal{F}_t)$ is an increasing family of σ-algebras representing information available at time t. A process M_t adapted to it, with $\mathbb{E}|M_t| < \infty$, is a *martingale* if $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ for $s \le t$.

Example 2.10 (Counting-process martingale)**.** For a homogeneous Poisson process of rate λ with its natural filtration, $M_t = N(t) - \lambda t$ is a martingale: for $s < t$, $\mathbb{E}[N(t) - N(s) \mid \mathcal{F}_s] = \lambda(t-s)$ by independent increments. More generally $N(t) - \int_0^t \lambda(u \mid \mathcal{H}_{u-})\,du$ is a (local) martingale for the conditional intensity, which is Result 74.

A *stopping time* τ is a random time such that $\{\tau \le t\} \in \mathcal{F}_t$: whether it has happened is known from the past. The first spike after stimulus onset is a stopping time; "the spike that precedes the largest ripple of the session" is not.

Theorem 2.11 (Optional stopping, bounded case)**.** *If M is a martingale and τ a bounded stopping time, then $\mathbb{E}M_\tau = \mathbb{E}M_0$.*

Proof in discrete time. If $\tau \le K$, write $M_\tau = M_0 + \sum_{k=1}^{K} (M_k - M_{k-1})\mathbf{1}_{\{\tau \ge k\}}$. The event $\{\tau \ge k\} = \{\tau \le k-1\}^c$ belongs to $\mathcal{F}_{k-1}$, so each term has expectation $\mathbb{E}[\mathbf{1}_{\{\tau \ge k\}}\mathbb{E}(M_k - M_{k-1} \mid \mathcal{F}_{k-1})] = 0$. □

Applied to $N(t) - \lambda t$, optional stopping gives $\mathbb{E}N(\tau) = \lambda \mathbb{E}\tau$ for bounded stopping times: the expected number of spikes before a history-dependent stopping rule equals the rate times the expected duration, as long as the rule does not look into the future. Analyses that end an epoch "at the last spike" violate this and are biased.

Exercises for this chapter

Exercise 2.12. Show that the intersection of two σ-algebras is a σ-algebra but their union need not be.

Exercise 2.13. Show that convergence in distribution to a constant implies convergence in probability to that constant.

Exercise 2.14. For a Poisson process, show that $(N(t) - \lambda t)^2 - \lambda t$ is a martingale.

Exercise 2.15. Decide whether "the first time the LFP envelope exceeds five MADs" and "the time of the session's largest envelope value" are stopping times.

Chapter 3

Real analysis, inequalities, and integration

Before manipulating signals, one must specify their domains, continuity, integrability, and normed spaces. A sampled waveform is a finite vector; a continuous-time waveform belongs to a function class. The two are connected through sampling hypotheses, not merely by notation.

Result 11: Triangle inequality. *Precise setting:* In a normed space $\|x + y\| \leq \|x\| + \|y\|$.

Why it matters here: Bound approximation and reconstruction errors.

Result 12: Cauchy–Schwarz inequality. *Precise setting:* In an inner-product space $|\langle x, y\rangle| \leq \|x\|\|y\|$; for square-integrable random variables, $|\operatorname{Cov}(X, Y)|^2 \leq \operatorname{Var}(X) \operatorname{Var}(Y)$.

Why it matters here: Bound Pearson correlation, coherence, and inner-product decoders.

Result 13: Holder's inequality. *Precise setting:* For $1/p + 1/q = 1$ with $p, q \geq 1$, and $f \in L^p$, $g \in L^q$, $\int |fg| \leq \|f\|_p \|g\|_q$.

Why it matters here: Establish existence of products and moments.

Result 14: Minkowski's inequality. *Precise setting:* For $p \geq 1$, $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.

Why it matters here: Justify L^p norms on signals and residuals.

Result 15: Bolzano–Weierstrass theorem. *Precise setting:* Every bounded sequence in finite-dimensional Euclidean space has a convergent subsequence.

Why it matters here: Understand compactness arguments in parameter estimation.

Result 16: Heine–Borel theorem. *Precise setting:* A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

Why it matters here: Provide conditions for existence of finite-dimensional constrained optima.

Result 17: Extreme value theorem. *Precise setting:* Continuous functions on nonempty compact sets attain their minimum and maximum.

Why it matters here: Show that certain constrained model-fitting objectives have minimizers.

Result 18: Mean value theorem. *Precise setting:* If f is continuous on $[a, b]$ and differentiable on (a, b), then some $c \in (a, b)$ satisfies $f(b) - f(a) = f'(c)(b - a)$.

Why it matters here: Control local approximation and numerical differentiation errors.

Result 19: Taylor's theorem with remainder. *Precise setting:* Under requisite differentiability, a function equals its order-k Taylor polynomial plus a controlled remainder.

Why it matters here: Linearize nonlinear neural state equations near operating points.

Result 20: Fundamental theorem of calculus. *Precise setting:* For continuous f, $F(x) = \int_a^x f(t)\,dt$ satisfies $F'(x) = f(x)$; suitable primitives recover integrals.

Why it matters here: Relate firing intensities to integrated expected event counts.

Result 21: Dominated convergence theorem. *Precise setting:* If measurable $f_n \to f$ almost everywhere and $|f_n| \le g$ for integrable g, then $\int f_n \to \int f$.

Why it matters here: Justify exchanging controlled limits and expectations.

Result 22: Monotone convergence and Fatou's lemma. *Precise setting:* For nonnegative measurable increasing $f_n \uparrow f$, $\int f_n \uparrow \int f$; Fatou gives $\int \liminf f_n \le \liminf \int f_n$.

Why it matters here: Handle nonnegative measures and event-count approximations.

Result 23: Radon–Nikodym theorem. *Precise setting:* If ν is a sigma-finite measure absolutely continuous relative to a sigma-finite measure μ, there is a measurable density $d\nu/d\mu$.

Why it matters here: Define likelihood and probability density correctly.

Result 24: Banach fixed-point theorem. *Precise setting:* A contraction on a nonempty complete metric space has exactly one fixed point, reached by iteration from any starting point.

Why it matters here: Justify convergence of certain recursive estimators under contraction.

Definition 3.1. A sequence x_n converges to x in a metric space if for every $\epsilon > 0$ there is an N such that $n \ge N$ implies $d(x_n, x) < \epsilon$. Convergence in probability, almost surely, and in L^2 have different definitions; none should silently replace another.

Interpretation warning 3.2. The derivative of an estimated signal can amplify measurement noise. A valid derivative theorem for a smooth function does not prove that a finite-difference estimate of noisy LFP samples is accurate.

3.1 Proofs of selected inequalities

Lemma 3.3 (Young's inequality)**.** *For $a, b \ge 0$ and $p, q > 1$ with $1/p + 1/q = 1$, $ab \le a^p/p + b^q/q$.*

Proof. If $ab = 0$ there is nothing to prove. Otherwise, by concavity of the logarithm, $\log(a^p/p + b^q/q) \ge \frac{1}{p}\log a^p + \frac{1}{q}\log b^q = \log(ab)$. □

Proof of Result 13 (Hölder). If $\|f\|_p = 0$ or $\|g\|_q = 0$ the product vanishes almost everywhere. Otherwise set $F = |f|/\|f\|_p$ and $G = |g|/\|g\|_q$. Young's inequality gives $FG \le F^p/p + G^q/q$ pointwise; integrating yields $\int FG \le 1/p + 1/q = 1$, which is the claim after rescaling. □

Proof of Result 14 (Minkowski). For $p = 1$ use the pointwise triangle inequality. For $p > 1$, $|f+g|^p \le |f||f+g|^{p-1} + |g||f+g|^{p-1}$. Apply Hölder with exponents p and $q = p/(p-1)$ to each term: $\|f+g\|_p^p \le (\|f\|_p + \|g\|_p)\,\|f+g\|_p^{p-1}$, and divide. □

Hölder with $p = q = 2$ is Cauchy–Schwarz. On a probability space, Hölder applied to $|X|^r \cdot 1$ with exponent s/r gives *Lyapunov's inequality* $(\mathbb{E}|X|^r)^{1/r} \le (\mathbb{E}|X|^s)^{1/s}$ for $0 < r < s$. In particular a finite fourth moment guarantees finite second and third moments, which is the hypothesis needed before reporting skewness and kurtosis of envelope amplitudes.

3.2 Contractions and fixed points

Proof of Result 24 (Banach). Let T satisfy $d(Tx, Ty) \le c\, d(x, y)$ with $0 \le c < 1$ on a complete metric space. For any x_0, the iterates $x_{n+1} = Tx_n$ satisfy $d(x_{n+1}, x_n) \le c^n d(x_1, x_0)$, so for $m > n$,

$$d(x_m, x_n) \le \sum_{k=n}^{m-1} c^k d(x_1, x_0) \le \frac{c^n}{1-c} d(x_1, x_0).$$

The sequence is Cauchy and converges to some x^*; continuity of T gives $Tx^* = x^*$. If $Ty = y$ as well, then $d(x^*, y) \le c\, d(x^*, y)$ forces $y = x^*$. □

The proof yields an a-priori error bound $d(x_n, x^*) \le c^n d(x_1, x_0)/(1-c)$, which is a usable stopping rule when c is known. Fixed-point iterations in spike-sorting template matching, in self-consistent mean-field rate models, and in the existence proof for ODE solutions (Result 81) all rely on this theorem.

Example 3.4 (Mean-field firing rate)**.** In a recurrent population with rate $r = \phi(wr + I)$ and a transfer function with Lipschitz constant L_ϕ, the map $T(r) = \phi(wr + I)$ is a contraction when $|w| L_\phi < 1$. A unique self-consistent rate then exists and the iteration $r_{n+1} = T(r_n)$ finds it. When $|w| L_\phi \ge 1$ multiple fixed points (bistability) or no stable fixed point can occur; the theorem is silent in that regime.

3.3 Likelihoods are Radon–Nikodym derivatives

Result 23 is what makes "the likelihood" a well-defined object across discrete, continuous, and point-process data. A family of probability measures P_θ is *dominated* by a reference measure ν if every $P_\theta \ll \nu$; the likelihood is then $L(\theta; x) = \frac{dP_\theta}{d\nu}(x)$. Counting measure gives probability mass functions, Lebesgue measure gives densities, and for spike trains on $[0, T]$ the reference is the law of a unit-rate Poisson process, which yields

$$\frac{dP_\lambda}{dP_1}(t_1, \dots, t_n) = \exp\Big(\sum_{i=1}^{n} \log \lambda(t_i) - \int_0^T (\lambda(t) - 1)\, dt\Big),$$

matching Result 76 up to a parameter-free factor. Changing the reference measure multiplies every likelihood by the same function of the data, so likelihood ratios and maximum-likelihood estimates are unaffected.

3.4 Exchanging limits, sums, and integrals in practice

Three standard tools justify most interchanges in this book:

1. *Monotone convergence* for nonnegative increasing integrands;
2. *Dominated convergence* with an integrable bound;
3. *Fubini–Tonelli* for iterated integrals and sums, with absolute integrability or nonnegativity.

Example 3.5 (Differentiating under the integral)**.** Let $\ell(\theta) = \int \log p_\theta(x)\, dP(x)$. If $|\partial_\theta \log p_\theta(x)| \le g(x)$ for θ near θ_0 with $\int g\, dP < \infty$, the mean value theorem and dominated convergence give

$$\ell'(\theta_0) = \int \partial_\theta \log p_\theta(x)\big|_{\theta=\theta_0}\, dP(x).$$

This is the step behind "the expected score is zero" in the proof of the Cramér–Rao inequality (Result 50).

Exercises for this chapter

Exercise 3.6. Show that equality holds in Hölder's inequality iff $|f|^p$ and $|g|^q$ are proportional almost everywhere.

Exercise 3.7. Find the contraction constant of $T(x) = \cos x$ on $[0, 1]$ and the number of iterations needed to approximate its fixed point to 10^{-6} using the a-priori bound.

Exercise 3.8. Give an example in which Fatou's inequality is strict.

Exercise 3.9. Derive the Poisson-process likelihood ratio above from the binned Bernoulli likelihood by letting the bin width tend to zero.

Part II

Vector spaces and statistical inference

Chapter 4

Vector spaces, matrices, and high-dimensional geometry

Let $X \in \mathbb{R}^{T \times N}$ collect N neural variables across T observations. Its columns may be neurons, channels, or features; centering across rows changes the statistical quantity being modeled. Linear algebra does not specify how trials should be aligned or how observations should be validated.

Result 25: Rank–nullity theorem. *Precise setting:* For linear $A : V \to W$ with $\dim V < \infty$, $\dim \ker A + \operatorname{rank} A = \dim V$.

Why it matters here: Count identifiable and unidentifiable latent directions.

Result 26: Orthogonal projection theorem. *Precise setting:* In a Hilbert space every element has a unique closest point in a closed subspace; the residual is orthogonal to that subspace.

Why it matters here: Derive least squares and orthogonal signal decompositions.

Result 27: Spectral theorem. *Precise setting:* A real symmetric matrix admits an orthonormal eigenbasis and real eigenvalues; a covariance matrix has nonnegative eigenvalues.

Why it matters here: Explain the eigenvectors underlying population PCA.

Result 28: Singular value decomposition. *Precise setting:* Every real matrix X admits $X = U\Sigma V^\top$ with orthogonal factors and nonnegative singular values.

Why it matters here: Implement low-rank population representations.

Result 29: Eckart–Young–Mirsky theorem. *Precise setting:* Truncating SVD to rank k gives an optimal rank-k approximation in Frobenius and spectral norms, more generally unitarily invariant norms.

Why it matters here: Give a reconstruction criterion for reduced-dimensional activity.

Result 30: Courant–Fischer min–max theorem. *Precise setting:* Eigenvalues of a symmetric matrix have variational characterizations over subspaces via Rayleigh quotients.

Why it matters here: Derive why sequential PCA directions maximize remaining variance.

Result 31: Gauss–Markov theorem. *Precise setting:* With a full-column-rank linear model, zero-mean errors with common variance and no correlation, OLS is BLUE.

Why it matters here: Clarify which optimality claim linear decoders can and cannot make.

Result 32: Schur complement identities. *Precise setting:* For an invertible block, Schur complements characterize positive definiteness and block inverses under the stated conditions.

Why it matters here: Compute conditional Gaussian covariance and latent uncertainty.

Result 33: Weyl eigenvalue inequalities. *Precise setting:* For Hermitian A, E, sorted eigenvalues of $A + E$ differ from those of A by at most $\|E\|_2$.

Why it matters here: Control eigenspectrum changes caused by covariance perturbations.

Result 34: Davis–Kahan sin-theta theorem. *Precise setting:* For symmetric matrices with separated spectral clusters, their invariant subspaces have angle bounds proportional to perturbation size divided by a suitable eigengap.

Why it matters here: Assess when estimated neural subspaces are stable.

Result 35: Sherman–Morrison–Woodbury identity. *Precise setting:* For invertible A, C and required inverse, $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$.

Why it matters here: Speed covariance updates when their rank is small.

Result 36: Matrix determinant lemma. *Precise setting:* For invertible A, C, $\det(A + UCV) = \det(C^{-1} + VA^{-1}U)\det(C)\det(A)$.

Why it matters here: Accelerate Gaussian latent-variable likelihood evaluations.

Result 37: Mercer's theorem. *Precise setting:* For a continuous positive-semidefinite kernel on a compact domain with a suitable finite reference measure, an orthogonal eigenfunction expansion exists under its hypotheses.

Why it matters here: Understand kernel feature representations in GP modeling.

Result 38: Karhunen–Loeve theorem. *Precise setting:* A second-order process admits a mean-square orthogonal expansion in covariance eigenfunctions under suitable regularity.

Why it matters here: Link temporal covariance to latent trajectories and function-space PCA.

Result 39: Kruskal's uniqueness criterion for CP tensors. *Precise setting:* A canonical polyadic tensor decomposition is essentially unique if its factor Kruskal ranks meet an appropriate sufficient bound, e.g. $k_A + k_B + k_C \geq 2R + 2$ for three factors.

Why it matters here: Reason about identifiability in trial-by-time-by-neuron decompositions.

Definition 4.1. For covariance $C \succeq 0$ with eigenvalues λ_i, the participation ratio is

$$D_{\mathrm{PR}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

It measures effective spectral dimensionality, not the intrinsic manifold dimension without further assumptions.

Interpretation warning 4.2. Truncated SVD solves a matrix approximation problem; recovering biological latent variables is an additional inference problem. Coherent noise, trial structure, and preprocessing can dominate the covariance eigenspectrum.

4.1 Proofs of selected matrix results

Proof of Result 31 (Gauss–Markov). Let $y = X\beta + \varepsilon$ with $X \in \mathbb{R}^{T\times p}$ of full column rank, $\mathbb{E}\varepsilon = 0$ and $\operatorname{Cov}\varepsilon = \sigma^2 I$. The OLS estimator is $\hat\beta = (X^\top X)^{-1}X^\top y$, with $\operatorname{Cov}\hat\beta = \sigma^2(X^\top X)^{-1}$. Any linear estimator is $\tilde\beta = Ay$; unbiasedness for every β requires $AX = I$. Write $A = (X^\top X)^{-1}X^\top + D$; then $DX = 0$, and

$$\operatorname{Cov}\tilde\beta = \sigma^2 AA^\top = \sigma^2(X^\top X)^{-1} + \sigma^2 DD^\top,$$

because the cross terms contain $DX = 0$. Since $DD^\top \succeq 0$, $\operatorname{Cov}\tilde\beta - \operatorname{Cov}\hat\beta \succeq 0$. □

The theorem's hypotheses are restrictive in neural data. Binned spike counts are heteroskedastic (their variance grows with their mean) and time bins are autocorrelated; under either violation OLS remains unbiased but is no longer best, and its textbook standard errors are wrong. Weighted or generalized least squares, with $\operatorname{Cov}\varepsilon = \sigma^2\Omega$, restores optimality by whitening with $\Omega^{-1/2}$.

Proof of Result 32 (Schur complement). Let $M = \left(\begin{smallmatrix} A & B \\ B^\top & C \end{smallmatrix}\right)$ be symmetric with $C \succ 0$, and $S = A - BC^{-1}B^\top$. The block factorization

$$M = \begin{pmatrix} I & BC^{-1} \\ 0 & I \end{pmatrix}\begin{pmatrix} S & 0 \\ 0 & C \end{pmatrix}\begin{pmatrix} I & 0 \\ C^{-1}B^\top & I \end{pmatrix}$$

is verified by multiplication. The outer factors are invertible and transposes of each other, so M is congruent to $\operatorname{diag}(S, C)$. Hence $M \succ 0$ iff $S \succ 0$ (given $C \succ 0$), $\det M = \det S \det C$, and inverting the three factors gives the block inverse, whose top-left block is S^{-1}. □

In Gaussian conditioning (Chapter 6), S is exactly the conditional covariance. The determinant identity $\det M = \det S \det C$ is the chain rule of Gaussian likelihoods: $\log p(x, y) = \log p(x \mid y) + \log p(y)$.

Proof of Result 35 (Woodbury). Multiply $A + UCV$ by the proposed inverse. The product is

$$I - U(C^{-1} + VA^{-1}U)^{-1}VA^{-1} + UCVA^{-1} - UCVA^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}.$$

Collect the terms ending in $(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$: their coefficient is $-U - UCVA^{-1}U = -UC(C^{-1} + VA^{-1}U)$. Thus those terms equal $-UCVA^{-1}$, which cancels the remaining $+UCVA^{-1}$. □

For factor analysis with $\Sigma = \Psi + \Lambda\Lambda^\top$, Ψ diagonal $N \times N$ and Λ of rank $k \ll N$, Woodbury reduces Σ^{-1} to the inversion of a $k \times k$ matrix: $\Sigma^{-1} = \Psi^{-1} - \Psi^{-1}\Lambda(I_k + \Lambda^\top\Psi^{-1}\Lambda)^{-1}\Lambda^\top\Psi^{-1}$. For $N = 1000$ neurons and $k = 10$ factors the cost drops from $O(N^3)$ to $O(Nk^2)$.

Proof of Result 33 (Weyl). By Courant–Fischer (Result 30),

$$\lambda_k(A + E) = \max_{\dim W = k} \min_{x \in W,\ \|x\|=1} x^\top(A + E)x.$$

For every unit vector x,

$$x^\top Ax - \|E\|_2 \le x^\top(A + E)x \le x^\top Ax + \|E\|_2.$$

Taking min over W then max over subspaces preserves the inequalities. □

4.2 Subspace stability in practice

Result 34 (Davis–Kahan) has a usable form: if $\hat{\Sigma} = \Sigma + E$ and the top-k eigenspace of Σ is separated from the rest by the gap $\delta = \lambda_k - \lambda_{k+1} > 0$, then the principal angles Θ between the true and estimated top-k subspaces satisfy

$$\|\sin\Theta\|_2 \le \frac{2\|E\|_2}{\delta}.$$

For a sample covariance of T observations of N roughly Gaussian variables, $\|E\|_2$ is of order $\|\Sigma\|_2(\sqrt{N/T} + N/T)$. Two consequences follow. First, principal subspaces are well estimated only when the eigengap exceeds the sampling fluctuation. Second, individual components inside a cluster of nearly equal eigenvalues are not estimable, while the subspace spanned by the whole cluster may be.

Example 4.3 (Split-half reliability of PCs)**.** Split trials into two halves, compute PCA on each, and measure the principal angles between the top-k subspaces. Small angles for the subspace with large angles for individual PCs indicate rotation within a near-degenerate cluster: report the subspace, not the components. This is an empirical counterpart of the Davis–Kahan bound.

4.3 Karhunen–Loève expansions of two standard processes

Result 38 states that a mean-square continuous process on $[0, T]$ with covariance $K(s, t)$ admits $X(t) = \sum_j \sqrt{\lambda_j}\,\xi_j\, e_j(t)$ with uncorrelated unit-variance ξ_j, where (λ_j, e_j) are eigenpairs of the covariance operator.

Example 4.4 (Brownian motion)**.** For $K(s, t) = \min(s, t)$ on $[0, 1]$, the eigenproblem $\int_0^1 \min(s, t)e(s)\,ds = \lambda e(t)$ differentiates twice into $-e'' = e/\lambda$ with $e(0) = 0$, $e'(1) = 0$. Hence $e_j(t) = \sqrt{2}\sin((j - \frac{1}{2})\pi t)$ and $\lambda_j = 1/((j - \frac{1}{2})^2\pi^2)$. The eigenvalues decay as j^{-2}: most of the variance lies in a few slow components. A random walk analysed by PCA across time therefore produces smooth, sinusoid-like "components" that reflect only its covariance, not oscillations.

This example explains a well-known pitfall: PCA of smoothly drifting population activity yields leading components resembling sines and cosines of increasing frequency, which can be mistaken for rotational or oscillatory dynamics.

Exercises for this chapter

Exercise 4.5. Show that the OLS residual $y - X\hat{\beta}$ satisfies $X^\top(y - X\hat{\beta}) = 0$ and interpret this as Result 26.

Exercise 4.6. Prove the matrix determinant lemma (Result 36) from the Schur complement determinant identity applied to $\left(\begin{smallmatrix} A & -U \\ V & C^{-1} \end{smallmatrix}\right)$.

Exercise 4.7. For $\Sigma = \mathrm{diag}(3, 2.9, 1)$ and a perturbation of norm 0.2, compare the Davis–Kahan bound for the top-one and top-two subspaces.

Exercise 4.8. Verify that $e_j(t) = \sqrt{2}\sin((j - \frac{1}{2})\pi t)$ are orthonormal on $[0, 1]$.

Chapter 5

Probability, estimation, and valid inference

A random variable is a measurable function on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. An expectation is an integral with respect to $\mathbb{P}$. The distinction between a deterministic signal and a stochastic process must be preserved even when both are stored as the same array of samples.

Result 40: Bayes' theorem and total probability. *Precise setting:* For events with positive conditioning probability, $\mathbb{P}(A \mid B) = \mathbb{P}(B \mid A)\mathbb{P}(A)/\mathbb{P}(B)$; for a partition (A_i), $\mathbb{P}(B) = \sum_i \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$.

Why it matters here: Build probabilistic decoders and posterior state estimates.

Result 41: Law of total expectation. *Precise setting:* For integrable Y, $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid G]]$.

Why it matters here: Marginalize trial-to-trial latent gain or unobserved state.

Result 42: Law of total variance. *Precise setting:* For square-integrable Y, $\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid G)] + \mathrm{Var}(\mathbb{E}[Y \mid G])$.

Why it matters here: Explain Poisson noise plus variable excitability.

Result 43: Law of total covariance. *Precise setting:* For square-integrable X, Y, $\mathrm{Cov}(X, Y) = \mathbb{E}[\mathrm{Cov}(X, Y \mid G)] + \mathrm{Cov}(\mathbb{E}[X \mid G], \mathbb{E}[Y \mid G])$.

Why it matters here: Separate residual noise correlation from shared modulating factors.

Result 44: Law of large numbers. *Precise setting:* Sample means converge to the expectation under appropriate independence or dependence, moment, and ergodicity assumptions.

Why it matters here: Explain consistency and why many time bins need not imply many independent samples.

Result 45: Central limit theorem. *Precise setting:* For iid observations with finite nonzero variance, normalized sample-mean fluctuations converge in distribution to a normal law; dependent-data versions need stronger conditions.

Why it matters here: Motivate uncertainty quantification for population statistics.

Result 46: Birkhoff ergodic theorem. *Precise setting:* For an integrable observable of a

measure-preserving system, time averages converge almost surely to a conditional invariant expectation; ergodicity makes the limit equal the ensemble expectation.

Why it matters here: Clarify when one long recording can replace independent replicates.

Result 47: Markov and Chebyshev inequalities. *Precise setting:* For $X \geq 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$; for finite variance, $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \mathrm{Var}(X)/a^2$.

Why it matters here: Derive conservative uncertainty bounds for estimators.

Result 48: Hoeffding and Bernstein concentration bounds. *Precise setting:* Bound sums of independent bounded variables, or those obeying specified moment constraints; extensions to dependence require care.

Why it matters here: Control finite-sample false alarms under valid independence assumptions.

Result 49: Delta method. *Precise setting:* A differentiable map of an asymptotically normal estimator is asymptotically normal after suitable scaling.

Why it matters here: Approximate uncertainty of nonlinear indices.

Result 50: Cramer–Rao inequality. *Precise setting:* With regular parametric likelihood and unbiased estimator, variance is bounded by reciprocal Fisher information (or its multivariate matrix counterpart).

Why it matters here: Compare theoretical accuracy limits of firing-rate estimators.

Result 51: Neyman–Pearson lemma. *Precise setting:* For testing one simple hypothesis against another, thresholding the likelihood ratio yields a most-powerful test at fixed size, allowing randomization at a threshold.

Why it matters here: Understand what an optimal detection test does and does not guarantee.

Result 52: Wilks' theorem. *Precise setting:* Under regularity and appropriate interior-point assumptions, twice a nested-model log-likelihood difference is asymptotically chi-squared with degrees of freedom equal to parameter difference.

Why it matters here: Test nested GLM or autoregressive restrictions when assumptions apply.

Result 53: Fisher–Neyman factorization theorem. *Precise setting:* In dominated statistical families, a statistic is sufficient precisely when the likelihood has an appropriate factorization.

Why it matters here: Justify reductions of data into counts only for particular models.

Result 54: Exact randomization testing under exchangeability. *Precise setting:* If the null distribution is invariant under a finite transformation group, a correctly ranked test statistic gives a valid finite-sample randomization test.

Why it matters here: Build scientifically matched permutation and jitter controls.

Result 55: Benjamini–Hochberg FDR control. *Precise setting:* For valid p-values under independence or certain positive dependence assumptions, the step-up procedure controls expected false discovery proportion.

Why it matters here: Correct across electrodes and spectral comparisons.

Result 56: Isserlis' or Wick's Gaussian moment theorem. *Precise setting:* Centered jointly Gaussian even-order moments decompose into sums over pairings of covariances; odd-order moments vanish.

Why it matters here: Evaluate Gaussian baseline moments and covariance estimators.

Example 5.1 (A modulated Poisson count)**.** Let $N \mid G \sim \text{Poisson}(\mu G)$, $G > 0$, and $\mathbb{E}G = 1$. The law of total variance gives

$$\mathbb{E}N = \mu, \qquad \text{Var}(N) = \mu + \mu^2 \,\text{Var}(G), \qquad \frac{\text{Var}(N)}{\mathbb{E}N} = 1 + \mu \,\text{Var}(G).$$

This derivation explains overdispersion without claiming that a particular dataset actually follows this latent-gain model. See Goris et al. [13].

Interpretation warning 5.2. A confidence interval does not automatically have its nominal coverage after choosing frequency bands, time windows, and hypotheses from the same data without accounting for selection.

5.1 Proofs of selected inference results

Proof of Result 43 (total covariance). Write $X - \mathbb{E}X = (X - \mathbb{E}[X \mid G]) + (\mathbb{E}[X \mid G] - \mathbb{E}X)$ and similarly for Y. Expanding the product, the two cross terms vanish because $\mathbb{E}[(X - \mathbb{E}[X \mid G])\, h(G)] = 0$ for any square-integrable $h(G)$. The remaining terms are $\mathbb{E}[\text{Cov}(X, Y \mid G)]$ and $\text{Cov}(\mathbb{E}[X \mid G], \mathbb{E}[Y \mid G])$. □

For two neurons modulated by a common gain G with $X \mid G \sim \text{Poisson}(\mu_1 G)$, $Y \mid G \sim \text{Poisson}(\mu_2 G)$, conditionally independent, the first term vanishes and $\text{Cov}(X, Y) = \mu_1 \mu_2 \,\text{Var}\, G$. Shared gain alone produces positive noise correlations proportional to the product of the rates, which is one reason correlations scale with firing rate.

Proof of Result 50 (Cramér–Rao, scalar case). Let $U = \partial_\theta \log p_\theta(X)$. Under conditions allowing differentiation under the integral, $\mathbb{E}_\theta U = 0$ and, for any unbiased $\hat{\theta}$, $\partial_\theta \mathbb{E}_\theta \hat{\theta} = 1 = \mathbb{E}_\theta[\hat{\theta} U] = \text{Cov}_\theta(\hat{\theta}, U)$. Cauchy–Schwarz gives $1 \le \text{Var}_\theta(\hat{\theta})\ \text{Var}_\theta(U) = \text{Var}_\theta(\hat{\theta})\, \mathcal{I}(\theta)$. □

For n independent Poisson counts in windows of length Δ with rate r, $\mathcal{I}(r) = n\Delta/r$, so no unbiased estimator of the rate has variance below $r/(n\Delta)$, which the pooled rate attains. More recording time, not finer binning, is the only way to improve it.

Example 5.3 (Result 53: counts are sufficient for Poisson rates)**.** For a homogeneous Poisson process on $[0, T]$, the likelihood of spike times $t_1, \ldots, t_n$ is $\lambda^n e^{-\lambda T}$. It depends on the data only through n, so the count is sufficient for λ: spike timing carries no further information about the rate *under this model*. For an inhomogeneous or history-dependent model, the likelihood depends on the times, and binning to counts discards information. Sufficiency is always relative to a model.

5.2 The delta method in neural statistics

If $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow \mathcal{N}(0, \sigma^2)$ and g is differentiable at θ with $g'(\theta) \neq 0$, then $\sqrt{n}(g(\hat{\theta}) - g(\theta)) \Rightarrow \mathcal{N}(0, g'(\theta)^2 \sigma^2)$.

Proof sketch. By Taylor's theorem $g(\hat\theta) - g(\theta) = g'(\tilde\theta)(\hat\theta - \theta)$ with $\tilde\theta$ between the two; $\tilde\theta \to \theta$ in probability, so Slutsky's lemma applies. $\square$

Example 5.4 (Standard error of power in decibels)**.** If a spectral estimate is $\hat S \approx S\chi^2_{2K}/(2K)$, then $\operatorname{Var}\hat S \approx S^2/K$. With $g(S) = 10\log_{10} S$, $g'(S) = 10/(S\ln 10)$ and $\operatorname{sd}(10\log_{10}\hat S) \approx 10/(\ln 10\sqrt{K}) \approx 4.34/\sqrt{K}\,\mathrm{dB}$, independent of S. The log transform stabilizes the variance, which is why spectra and their condition differences are analysed in decibels.

Example 5.5 (Fano factor)**.** For n counts with mean μ, variance v and Fano factor $F = v/\mu$, the multivariate delta method gives an approximate standard error that grows as the counts become more skewed. A simple bootstrap over trials is often more reliable than the asymptotic formula when $n < 50$.

5.3 Concentration: Hoeffding's inequality

Theorem 5.6 (Hoeffding)**.** *If $X_1, \dots, X_n$ are independent with $a_i \le X_i \le b_i$, then for $t > 0$,*

$$\mathbb{P}(\bar X - \mathbb{E}\bar X \ge t) \le \exp\!\Big(-\frac{2n^2t^2}{\sum_i (b_i - a_i)^2}\Big).$$

For a decoder evaluated on n independent test trials with 0/1 loss, the observed accuracy is within t of its expectation with probability at least $1 - 2e^{-2nt^2}$. With $n = 100$ trials and $t = 0.1$ this is 0.73; to guarantee ± 0.05 with probability 0.95 requires $n \ge 738$. The guarantee assumes independent test trials; neighbouring time bins from the same trial are not.

5.4 Isserlis' theorem and the variance of covariance estimates

For a centered Gaussian vector, $\mathbb{E}[X_1X_2X_3X_4] = \Sigma_{12}\Sigma_{34} + \Sigma_{13}\Sigma_{24} + \Sigma_{14}\Sigma_{23}$. Applied to the product X_iX_j, it gives $\operatorname{Var}(X_iX_j) = \Sigma_{ii}\Sigma_{jj} + \Sigma_{ij}^2$, so the sample covariance from T independent samples has variance $(\Sigma_{ii}\Sigma_{jj} + \Sigma_{ij}^2)/T$. Translated into correlations, the standard error of a near-zero correlation is about $1/\sqrt{T}$: detecting pairwise correlations of 0.02 at two standard errors needs $T \approx 10^4$ independent samples.

Exercises for this chapter

Exercise 5.7. Derive the Fisher information of an exponential interspike-interval model and the Cramér–Rao bound for its rate.

Exercise 5.8. Use the delta method to find the approximate variance of $\log(\hat r_1/\hat r_2)$ for two independent Poisson rate estimates.

Exercise 5.9. Prove Isserlis' formula for four jointly Gaussian variables by differentiating the moment generating function $\exp(\tfrac12 s^\top \Sigma s)$.

Exercise 5.10. Show by an example that pairwise-independent Bernoulli variables can violate Hoeffding's bound for their mean.

Chapter 6

A working toolkit for Gaussian models

Gaussian conditioning silently appears in Kalman filters, Gaussian processes, factor analysis, and several multivariate estimators. It belongs before relying on their formulas.

Definition 6.1 (Multivariate normal vector)**.** A vector $X \in \mathbb{R}^d$ is jointly Gaussian if every scalar linear combination $a^\top X$ is normally distributed (possibly degenerate). Its mean μ and covariance $\Sigma \succeq 0$ characterize its distribution. When $\Sigma \succ 0$, its density is

$$p(x) = (2\pi)^{-d/2}|\Sigma|^{-1/2}\exp\left[-\frac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu)\right].$$

Theorem 6.2 (Gaussian conditioning by a Schur complement)**.** *Suppose (X, Y) is jointly normal and partition its mean and covariance into μ_X, μ_Y and blocks Σ_{XX}, Σ_{XY}, Σ_{YY}. If Σ_{YY} is invertible,*

$$\mathbb{E}[X \mid Y = y] = \mu_X + \Sigma_{XY}\Sigma_{YY}^{-1}(y - \mu_Y),$$
$$\mathrm{Cov}(X \mid Y) = \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}.$$

Proof. Define $R = X - \mu_X - \Sigma_{XY}\Sigma_{YY}^{-1}(Y - \mu_Y)$. Then $\mathrm{Cov}(R, Y) = 0$. Joint Gaussianity makes R and Y independent. Expanding $\mathrm{Cov}(R)$ gives the Schur complement. □

Proposition 6.3 (Wishart sample-covariance model)**.** *If $Z_1, \dots, Z_T$ are independent $\mathcal{N}_d(0, \Sigma)$ with known zero mean, then $\sum_{t=1}^{T} Z_t Z_t^\top$ has a Wishart distribution with scale Σ and T degrees of freedom. The usual centered sample covariance instead involves $T - 1$ degrees of freedom for independent Gaussian samples.*

This distributional assumption is the model behind classical random-covariance results; the same formula does not describe arbitrarily correlated neuronal time bins.

Proposition 6.4 (Multivariate Stein identity)**.** *For $X \sim \mathcal{N}_d(\mu, \Sigma)$ and a differentiable function f with appropriate integrability and boundary decay,*

$$\mathbb{E}[(X - \mu)f(X)] = \Sigma\,\mathbb{E}[\nabla f(X)].$$

Proof. For a nondegenerate Gaussian density p, $\nabla p(x) = -\Sigma^{-1}(x - \mu)p(x)$. Integrate $\nabla(fp)$ and use the vanishing boundary term. □

6.1 Gaussian KL divergence

For full-rank Gaussian distributions $P = \mathcal{N}(\mu_0, \Sigma_0)$ and $Q = \mathcal{N}(\mu_1, \Sigma_1)$ in dimension d,

$$D_{\mathrm{KL}}(P\|Q) = \frac{1}{2}\left[\operatorname{tr}(\Sigma_1^{-1}\Sigma_0) + (\mu_1 - \mu_0)^\top \Sigma_1^{-1}(\mu_1 - \mu_0) - d + \log\frac{\det\Sigma_1}{\det\Sigma_0}\right].$$

A proof expands $\mathbb{E}_P[\log p(X) - \log q(X)]$, then uses the covariance trace identity. The formula requires positive-definite covariance matrices; singular distributions need measure-theoretic treatment of their supports.

Example 6.5 (Neural latent variables)**.** A Kalman measurement update is repeated conditioning of a joint Gaussian state–observation vector. Its familiar gain matrix is therefore a Schur-complement calculation, not a property of all nonlinear or non-Gaussian observation processes.

6.2 Linear-Gaussian models

Most Gaussian computations in this book are instances of one pattern.

Proposition 6.6 (Linear-Gaussian marginal and posterior)**.** *Let $z \sim \mathcal{N}(m, P)$ and $y \mid z \sim \mathcal{N}(Hz, R)$ with $P, R \succ 0$. Then $y \sim \mathcal{N}(Hm, HPH^\top + R)$ and*

$$z \mid y \sim \mathcal{N}\big(m + K(y - Hm),\ (I - KH)P\big), \qquad K = PH^\top(HPH^\top + R)^{-1}.$$

Proof. The vector (z, y) is jointly Gaussian with $\operatorname{Cov}(z, y) = PH^\top$ and $\operatorname{Cov}(y) = HPH^\top + R$. Apply the conditioning theorem of this chapter. □

The Kalman filter (Result 88) applies this proposition once per time step; factor analysis applies it with z the latent factors and y the population response; Gaussian-process regression applies it with z the function values at test points.

Information form. Writing $\Lambda = P^{-1}$ (the precision), the posterior precision is $\Lambda + H^\top R^{-1}H$ and the posterior mean solves $(\Lambda + H^\top R^{-1}H)\hat{z} = \Lambda m + H^\top R^{-1}y$. In this form independent observations add precision, which explains why combining neurons improves a population estimate and why perfectly redundant neurons (identical rows of H with correlated noise) add less than their number suggests.

6.3 Gaussian-process regression of a firing-rate function

Let $f \sim \mathcal{GP}(0, k)$ with squared-exponential kernel $k(t, t') = s^2 \exp(-(t - t')^2/(2\ell^2))$, observed with noise as $y_i = f(t_i) + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. For test times t_*, the proposition gives

$$\mathbb{E}[f_* \mid y] = K_*(K + \sigma^2 I)^{-1}y, \qquad \operatorname{Cov}(f_* \mid y) = K_{**} - K_*(K + \sigma^2 I)^{-1}K_*^\top.$$

The length scale ℓ plays the role of a smoothing bandwidth, and maximizing the marginal likelihood $\log p(y) = -\frac{1}{2}y^\top(K + \sigma^2 I)^{-1}y - \frac{1}{2}\log\det(K + \sigma^2 I) - \frac{n}{2}\log 2\pi$ chooses it from the data. The Gaussian observation model is only an approximation for spike counts; GPFA and related methods apply it to square-root or smoothed counts, while point-process GP methods require non-conjugate approximations.

6.4 Entropy and mutual information of Gaussians

A d-dimensional Gaussian has differential entropy $h = \frac{1}{2}\log\big((2\pi e)^d \det\Sigma\big)$. For jointly Gaussian (X, Y),

$$I(X;Y) = \tfrac{1}{2}\log\frac{\det\Sigma_{XX}\det\Sigma_{YY}}{\det\Sigma},$$

and in the scalar case $I = -\frac{1}{2}\log(1-\rho^2)$. A correlation of 0.1 carries 0.005 nats; 0.9 carries 0.83 nats. For a Gaussian input sent through an additive-noise channel with a fixed noise covariance, Gaussian noise minimizes the mutual information. Gaussian formulas computed from second-order statistics are therefore a lower bound in that additive-noise setting, but not a general bound on the information carried by nonlinear or non-Gaussian neural responses.

Example 6.7 (Information of a population code under Gaussian noise)**.** For $y = s\,u + \epsilon$ with scalar stimulus $s \sim \mathcal{N}(0, \sigma_s^2)$, tuning vector $u \in \mathbb{R}^N$, and $\epsilon \sim \mathcal{N}(0, \Sigma)$,

$$I(s;y) = \tfrac{1}{2}\log\big(1 + \sigma_s^2\, u^\top \Sigma^{-1} u\big).$$

The quantity $u^\top\Sigma^{-1}u$ is the linear Fisher information of the population. Correlations along u reduce it; correlations orthogonal to u do not. The formula follows from the matrix determinant lemma applied to $\det(\Sigma + \sigma_s^2 u u^\top)$.

Exercises for this chapter

Exercise 6.8. Derive the information-form posterior from the covariance-form posterior using the Woodbury identity.

Exercise 6.9. Show that two jointly Gaussian variables are independent iff they are uncorrelated, and give a non-Gaussian counterexample.

Exercise 6.10. Compute the Gaussian KL divergence between $\mathcal{N}(0,1)$ and $\mathcal{N}(\mu,\sigma^2)$ and examine its behaviour as $\sigma \to 0$.

Exercise 6.11. Derive $I(s;y)$ in the population-code example.

Part III

Physical and digital origins of neural measurements

Chapter 7

From membrane currents to extracellular electrodes

An extracellular electrode does not measure a neuron's voltage in isolation. It measures a difference of electric potential created by spatially distributed ionic currents, tissue geometry, conductivity, referencing, and acquisition electronics [5, 11].

7.1 Electrochemical driving forces

For an ideal dilute permeant ion with valence $z \neq 0$, equality of electrochemical potentials on the two sides of a membrane gives

$$E_{\text{ion}} = \frac{RT}{zF} \log \frac{c_{\text{out}}}{c_{\text{in}}}.$$

This is the Nernst equilibrium potential under its concentration and temperature assumptions. The Goldman–Hodgkin–Katz formula extends the constant-field model to multiple monovalent ionic species with permeability parameters; it is not equivalent to measuring those permeabilities from a single LFP. In the Hodgkin–Huxley description the sodium current is modeled as $I_{\text{Na}} = \bar{g}_{\text{Na}} m^3 h (V - E_{\text{Na}})$; gating variables obey voltage-dependent kinetic equations. The passive cable approximation has the schematic form

$$c_m \partial_t V(x,t) = g_a \partial_{xx} V(x,t) - g_m (V - E_L) + i_{\text{ext}}(x,t),$$

with coefficients per unit length and consistent units. Cable and membrane models determine transmembrane sources; they are not by themselves electrode forward models.

7.2 Quasistatic field equation

At electrophysiological frequencies and spatial scales where inductive effects are negligible, write $\mathbf{E} = -\nabla\phi$. Let impressed current $\mathbf{J}_p$ and Ohmic return current $-\boldsymbol{\sigma}\nabla\phi$ sum to a divergence-free total current. Current conservation implies

$$\nabla\cdot(\boldsymbol{\sigma}\nabla\phi) = \nabla\cdot\mathbf{J}_p, \quad \text{or} \quad -\nabla\cdot(\boldsymbol{\sigma}\nabla\phi) = q, \quad q = -\nabla\cdot\mathbf{J}_p.$$

The tensor $\boldsymbol{\sigma}$ may be anisotropic and spatially inhomogeneous. To solve the equation, the domain, conductivity, boundary conditions and electrode reference must be specified. For an

infinite isotropic medium, a point source of strength I located at $\mathbf{r}_0$ gives, away from that point,

$$\phi(\mathbf{r}) = \frac{I}{4\pi\sigma|\mathbf{r}-\mathbf{r}_0|}.$$

Superposition yields a discrete source model $\mathbf{v} = G\mathbf{s}$ when the forward physics is linear, with lead-field entries obtained by integrating the Green kernel over chosen source basis functions. A dipole arises as the leading far-field contribution when net source current cancels; line-source models integrate along a dendrite.

Proposition 7.1 (Uniqueness for fixed Dirichlet conditions)**.** *Let Ω be a bounded connected region and $\boldsymbol{\sigma}$ uniformly positive definite. If two weak solutions of the same elliptic equation have equal Dirichlet boundary values, they agree (almost everywhere).*

Proof. Their difference w satisfies the homogeneous equation and zero boundary values. Multiply by w, integrate by parts, and use the boundary condition: $\int_\Omega \nabla w \cdot \boldsymbol{\sigma}\nabla w\, dx = 0$. Positivity implies $\nabla w = 0$; the boundary forces $w = 0$. For pure Neumann conditions, uniqueness holds only up to an additive constant, with a compatibility condition on source flux. □

7.3 Why the inverse need not be unique

Suppose an electrode array measures $\mathbf{v} = G\mathbf{s}$. If $G\mathbf{h} = 0$ for nonzero $\mathbf{h}$, then $G(\mathbf{s}+\mathbf{h}) = G\mathbf{s}$. This is exact finite-dimensional nonidentifiability. In the continuum, smoothing operators can also be ill-conditioned or compact, so tiny measurement errors may cause enormous source errors even when uniqueness holds in a restricted model. *Hadamard well-posedness* requires existence, uniqueness and continuous dependence of solutions on data. A linear operator is *compact* if it maps bounded sets into sets with compact closure; its singular values in infinite-dimensional inverse problems can converge to zero. For singular system $Gv_j = \sigma_j u_j$ and data b, the formal inverse contains $\langle b, u_j\rangle/\sigma_j$ and magnifies noise when $\sigma_j \to 0$. A classical Picard-type condition controls the squared normalized coefficients for compatible data. Tikhonov regularization instead solves

$$\widehat{\mathbf{s}}_\lambda = \arg\min_{\mathbf{s}}\{\|G\mathbf{s}-\mathbf{v}\|_2^2 + \lambda\|\mathbf{s}\|_2^2\} = (G^\top G + \lambda I)^{-1}G^\top\mathbf{v}.$$

Example 7.2 (MATLAB and interpretation)**.** `field_potential_forward.m` receives G; it does *not* compute tissue conductivity or the Green kernel. `field_potential_inverse_tikhonov.m` returns one regularized source estimate, not a unique membrane-current pattern. `current_source_density.m` needs spatial geometry and differentiated potentials; references and volume conduction remain crucial.

7.4 The Goldman–Hodgkin–Katz equations

The constant-field model assumes that the electric field is uniform across a membrane of thickness d, that ions cross independently, and that the flux of each species obeys the Nernst–Planck equation $J = -D(\partial_x c + \frac{zF}{RT}c\,\partial_x\phi)$. With $u = zFV/(RT)$ and permeability $P = D\beta/d$ (β a partition coefficient), integrating across the membrane gives the *GHK current equation*

$$I = Pz^2\frac{F^2V}{RT}\,\frac{c_{\text{in}} - c_{\text{out}}e^{-u}}{1-e^{-u}}.$$

The current vanishes at $c_{\mathrm{in}} = c_{\mathrm{out}} e^{-u}$, i.e. at the Nernst potential, and is a nonlinear (rectifying) function of V when concentrations differ. Setting the *total* current of K^+, Na^+ and Cl^- to zero gives the *GHK voltage equation*

$$V_{\mathrm{rest}} = \frac{RT}{F} \log \frac{P_{\mathrm{K}}[K]_{\mathrm{o}} + P_{\mathrm{Na}}[Na]_{\mathrm{o}} + P_{\mathrm{Cl}}[Cl]_{\mathrm{i}}}{P_{\mathrm{K}}[K]_{\mathrm{i}} + P_{\mathrm{Na}}[Na]_{\mathrm{i}} + P_{\mathrm{Cl}}[Cl]_{\mathrm{o}}}.$$

Example 7.3 (Resting potential)**.** With $[K]_{\mathrm{o}} = 4$, $[K]_{\mathrm{i}} = 140$, $[Na]_{\mathrm{o}} = 145$, $[Na]_{\mathrm{i}} = 12$, $[Cl]_{\mathrm{o}} = 120$, $[Cl]_{\mathrm{i}} = 10\,\mathrm{mM}$, relative permeabilities $1 : 0.04 : 0.45$ and $RT/F \approx 26.7\,\mathrm{mV}$ at $37^{\circ}\mathrm{C}$, the numerator is $4 + 5.8 + 4.5 = 14.3$ and the denominator $140 + 0.48 + 54 = 194.5$, giving $V_{\mathrm{rest}} \approx 26.7 \ln(0.0735) \approx -70\,\mathrm{mV}$. The potassium Nernst potential is $26.7 \ln(4/140) \approx -95\,\mathrm{mV}$. The rest is depolarized relative to E_{K} because of the sodium and chloride terms.

The GHK equations are a model, not a law: they fail when channels are multiply occupied, when ions interact, or when the field is not uniform. Their ratio form also shows that only *relative* permeabilities are determined by a reversal potential, so a single measured reversal cannot fix three permeabilities.

7.5 The Hodgkin–Huxley system

The complete space-clamped model is

$$\begin{aligned} C_m \dot{V} &= -\bar{g}_{\mathrm{Na}} m^3 h (V - E_{\mathrm{Na}}) - \bar{g}_{\mathrm{K}} n^4 (V - E_{\mathrm{K}}) - g_L (V - E_L) + I_{\mathrm{ext}}, \\ \dot{x} &= \alpha_x(V)(1 - x) - \beta_x(V) x, \qquad x \in \{m, h, n\}. \end{aligned}$$

Each gate relaxes to $x_\infty(V) = \alpha_x/(\alpha_x + \beta_x)$ with time constant $\tau_x(V) = 1/(\alpha_x + \beta_x)$. Sodium activation m is fast, inactivation h and potassium activation n are slow. The spike arises because m jumps quickly after a depolarization, giving positive feedback through E_{Na}, while the slower decrease of h and increase of n terminate it. Reducing the model by setting $m = m_\infty(V)$ and exploiting the approximate relation $h + n \approx 0.8$ yields a two-dimensional system whose phase plane resembles FitzHugh–Nagumo; repetitive firing begins at a Hopf bifurcation of its equilibrium (Chapter 18).

For extracellular recording the essential output is the sum of transmembrane currents per unit area, $i_m = C_m \dot{V} + i_{\mathrm{ion}}$. By charge conservation along a neuron, the integral of i_m over the whole cell surface is zero at every instant: a neuron has no net monopole source, and its far field is at least dipolar.

7.6 Cable equation: space and time constants

With membrane resistance r_m and capacitance c_m per unit length, and axial resistance r_a per unit length, the passive cable is $\lambda^2 \partial_{xx} V - \tau \partial_t V - V = -r_m i_{\mathrm{ext}}$ with $\lambda = \sqrt{r_m / r_a}$ and $\tau = r_m c_m$. In steady state with a current injected at $x = 0$ of an infinite cable, $V(x) = V(0) e^{-|x|/\lambda}$. A sinusoidal input of angular frequency ω decays with the frequency-dependent length constant $\lambda_\omega = \lambda \sqrt{2/(1 + \sqrt{1 + \omega^2 \tau^2})}$, so dendrites filter high frequencies more strongly. Distal synaptic inputs therefore contribute low-pass-filtered dipoles, one physical reason why the LFP spectrum falls with frequency.

7.7 The Picard condition and regularization parameter choice

Let the discretized forward operator have SVD $G = \sum_j \sigma_j u_j v_j^\top$ and data $\mathbf{v} = G\mathbf{s}_{\mathrm{true}} + \boldsymbol{\epsilon}$. The *discrete Picard condition* requires that the coefficients $|u_j^\top \mathbf{v}|$ decay, on average, faster than σ_j until they reach the noise level. Plotting σ_j, $|u_j^\top \mathbf{v}|$ and their ratio on a logarithmic scale (a *Picard plot*) shows which components are informative: beyond the index where $|u_j^\top \mathbf{v}|$ levels off at the noise floor, the ratio grows and those components should be filtered. The Tikhonov estimate has filter factors $\sigma_j^2/(\sigma_j^2 + \lambda)$; common choices of λ are:

- the *discrepancy principle*: choose λ so that $\|G\hat{\mathbf{s}}_\lambda - \mathbf{v}\| \approx \|\boldsymbol{\epsilon}\|$, which requires an estimate of the noise level;
- the *L-curve*: plot $\log \|\hat{\mathbf{s}}_\lambda\|$ against $\log \|G\hat{\mathbf{s}}_\lambda - \mathbf{v}\|$ and choose the corner;
- *generalized cross-validation*, which minimizes $\|G\hat{\mathbf{s}}_\lambda - \mathbf{v}\|^2 / \operatorname{tr}(I - A_\lambda)^2$ with influence matrix $A_\lambda = G(G^\top G + \lambda I)^{-1} G^\top$.

Each criterion is a statistical decision with known failure modes: the L-curve can lack a corner, GCV can undersmooth with correlated noise, and the discrepancy principle depends on a noise estimate.

7.8 Line sources and laminar forward models

A neurite segment from $\mathbf{r}_1$ to $\mathbf{r}_2$ carrying a uniform transmembrane current I contributes

$$\phi(\mathbf{r}) = \frac{I}{4\pi\sigma\ell} \log \frac{\sqrt{h^2 + \rho^2} - h}{\sqrt{l^2 + \rho^2} - l},$$

where ℓ is the segment length, ρ the perpendicular distance of $\mathbf{r}$ from the segment's line, h the signed distance along the line from $\mathbf{r}$'s projection to $\mathbf{r}_2$, and $l = h + \ell$. This is the integral of the point-source kernel along the segment and is the building block of compartmental LFP simulators. It avoids the singularity of point sources near the electrode and shows that the potential of an elongated source decays logarithmically near it, much more slowly than $1/r$.

Exercises for this chapter

Exercise 7.4. For equal concentrations $c_{\mathrm{in}} = c_{\mathrm{out}} = c$, simplify the GHK current equation, taking the removable limit at $V = 0$, and show that $I = Pz^2F^2cV/(RT)$. Explain why unequal concentrations usually yield a voltage-dependent effective conductance and why a local linearization about a reversal potential is not a global Ohmic law.

Exercise 7.5. Compute the potassium, sodium and chloride Nernst potentials for the example concentrations.

Exercise 7.6. For $\tau = 20\,\mathrm{ms}$, compute λ_ω/λ at 1, 10 and 100 Hz.

Exercise 7.7. Derive the line-source formula by integrating $I\,ds/(4\pi\sigma\ell|\mathbf{r} - \mathbf{r}(s)|)$ along the segment.

Chapter 8

Digital filtering, stability, and aliasing

The first operation in many LFP workflows is band-pass filtering. Its interpretation requires a distinction between a sampled sequence, its discrete-time filter, and the underlying analog field [16].

8.1 Poisson summation and aliasing

For suitable rapidly decreasing $x(t)$, Fourier convention $X(f) = \int x(t)e^{-2\pi ift}\,dt$, and sampling frequency F_s, Poisson summation gives the periodic spectrum of its samples:

$$\sum_{n\in\mathbb{Z}} x(n/F_s)e^{-2\pi inf/F_s} = F_s \sum_{k\in\mathbb{Z}} X(f + kF_s).$$

The observed spectrum is thus a *sum of translated copies*. The copies overlap unless the analog spectrum is sufficiently bandlimited; an anti-alias filter must act *before* digitization. As a concrete counterexample, the sequences from frequencies f and $f + F_s$ agree because $e^{2\pi i(f+F_s)n/F_s} = e^{2\pi ifn/F_s}$. No post-acquisition algorithm can distinguish them without additional prior information.

8.2 Convolution, the z-transform, and poles

For a causal linear time-invariant discrete filter, $y[n] = \sum_{k=0}^{\infty} h[k]x[n-k]$, with

$$H(z) = \sum_{k=0}^{\infty} h[k]z^{-k}.$$

Its frequency response is $H(e^{i\omega})$ when that boundary value exists. Absolute summability $\sum_k |h[k]| < \infty$ is sufficient for bounded-input bounded-output (BIBO) stability. For rational causal filters in minimal form, BIBO stability requires all poles strictly inside the unit circle.

Proposition 8.1 (First-order recursive filter)**.** *For $y[n] = ay[n-1] + x[n]$ with zero initial condition, $h[k] = a^k\mathbf{1}_{k\geq 0}$. The system is BIBO stable if and only if $|a| < 1$.*

Proof. The absolute impulse-response sum is $\sum_{k\geq 0} |a|^k = (1 - |a|)^{-1}$ when $|a| < 1$. If $|a| \geq 1$, the impulse response is not absolutely summable; there exists a bounded input producing an unbounded output. □

8.3 FIR versus IIR; delay and zero-phase processing

An FIR filter has a finite impulse response and is BIBO stable. An IIR filter may have sharper transitions for a given order, but its poles and finite-precision implementation need inspection. For nonzero frequency response, define group delay $\tau_g(\omega) = -d \arg H(e^{i\omega})/d\omega$. A forward-backward filter ideally has response $H(e^{i\omega})H(e^{-i\omega}) = |H(e^{i\omega})|^2$ for real filter coefficients. This suppresses phase delay but is *noncausal*, changes the effective magnitude response, and requires boundary extrapolation: edge transients are not removed. An ideal brick-wall filter has an infinitely long sinc-shaped impulse response; abrupt truncation creates Gibbs-type ringing.

Example 8.2 (What the arsenal computes)**.** `bandpass_lfp.m` designs a Butterworth bandpass, converts it to second-order sections and applies `filtfilt`. The returned trace is a zero-phase *offline estimate*, not the output of a causal online instrument. The initial and final samples need explicit transient inspection. Filtering cannot remove volume conduction, spike-sorting errors or physiological common inputs.

8.4 Executable MATLAB experiment: two aliased sinusoids

The following script constructs a 15 Hz sinusoid and a 115 Hz sinusoid sampled at 100 Hz. Their continuous paths are different, but their sampled values coincide within floating-point tolerance. The code also exports vector PDF and raster PNG figures when run on a MATLAB-equipped workstation or self-hosted CI runner. The source below is included *directly* from the executable file; this book does not keep a separate printed copy.

```
% ALIASING_DEMO Reproducible synthetic sampling illustration.
% No toolbox, data file, graphics display or user workspace is required.
% MATLAB R2019a+; execution with run('examples/matlab/aliasing_demo.m').
fs = 100;                          % samples per second
duration_s = 0.20;
t = 0:1/10000:duration_s;          % finely resolved reference grid
n = 0:round(duration_s*fs);
f_low = 15;
f_alias = f_low + fs;
x_low = sin(2*pi*f_low*t);
x_high = sin(2*pi*f_alias*t);
samples_low = sin(2*pi*f_low*n/fs);
samples_high = sin(2*pi*f_alias*n/fs);
assert(max(abs(samples_low-samples_high)) < 1e-12, ...
    'Integer sampling should alias f and f+fs.');
fig = figure('Visible','off','Color','w');
plot(t,x_low,'LineWidth',1.4);
hold on;
plot(t,x_high,'LineWidth',0.8);
plot(n/fs,samples_low,'ko','MarkerFaceColor','k');
xlabel('Time (s)');
ylabel('Amplitude (a.u.)');
title('Two distinct continuous frequencies, identical sampled values');
legend('15 Hz','115 Hz','Samples','Location','best');
grid on;
destination = fullfile('books','electrophysiology','figures','generated');
if ~exist(destination,'dir'), mkdir(destination); end
exportgraphics(fig,fullfile(destination,'aliasing_demo.pdf'),'ContentType','vector');
exportgraphics(fig,fullfile(destination,'aliasing_demo.png'),'Resolution',180);
close(fig);
```

8.5 Linear-phase FIR filters

Proposition 8.3. *If a real FIR filter of length L has symmetric coefficients, $h[k] = h[L-1-k]$, then $H(e^{i\omega}) = e^{-i\omega(L-1)/2}A(\omega)$ with A real. Its group delay is the constant $(L-1)/2$ samples wherever $A(\omega) \neq 0$.*

Proof. Pair the terms k and $L-1-k$: $h[k](e^{-i\omega k}+e^{-i\omega(L-1-k)}) = 2h[k]e^{-i\omega(L-1)/2}\cos(\omega((L-1)/2-k))$. Summing gives the stated form with real A; the phase is linear with slope $-(L-1)/2$, except for jumps of π where A changes sign. □

A constant group delay means every frequency component is delayed by the same amount, so waveform shape is preserved up to a shift that can be removed exactly by shifting the output by $(L-1)/2$ samples. This gives a *causal* alternative to forward–backward filtering for offline analysis, with half the computational cost and without squaring the magnitude response. The price is that the shifted output depends on $(L-1)/2$ future samples, so it is not a real-time estimate.

8.6 Butterworth filters and the bilinear transform

An analog Butterworth low-pass filter of order n and cutoff Ω_c has $|H_a(i\Omega)|^2 = 1/(1 + (\Omega/\Omega_c)^{2n})$: it is maximally flat at $\Omega = 0$ and falls by 3 dB at the cutoff. Its poles lie equally spaced on the left half of the circle $|s| = \Omega_c$. A digital filter is obtained by the *bilinear transform* $s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$, which maps the left half-plane onto the unit disk, so stability is preserved. The map compresses the infinite analog frequency axis into $(-\pi, \pi)$ according to $\Omega = \frac{2}{T}\tan(\omega/2)$; cutoffs must be *pre-warped* by this relation before design.

Two numerical facts matter in practice. First, high-order IIR filters designed in transfer-function form (b, a) have polynomial coefficients whose roots are extremely sensitive to rounding, and narrow low-frequency band-passes at high sampling rates may become unstable or inaccurate; second-order sections (factoring into biquads) avoid this. Second, a band-pass Butterworth of design order n has $2n$ poles, and forward–backward application doubles the effective order and squares the magnitude response, so the -3 dB points of the designed filter become -6 dB points of the zero-phase result.

8.7 Notch filters and line noise

A second-order notch at normalized frequency ω_0 places zeros on the unit circle at $e^{\pm i\omega_0}$ and poles just inside at $re^{\pm i\omega_0}$:

$$H(z) = \frac{1 - 2\cos\omega_0 z^{-1} + z^{-2}}{1 - 2r\cos\omega_0 z^{-1} + r^2 z^{-2}}.$$

The closer r is to one, the narrower the notch (bandwidth about $(1-r)f_s/\pi$ Hz) and the longer its impulse response, approximately $1/(1-r)$ samples. A very narrow notch rings for a long time after a transient, which can produce 50 or 60 Hz artefacts after each large spike or stimulation pulse. Alternatives include regressing out line-frequency sinusoids estimated in sliding windows, or spectral interpolation, each with its own assumptions.

8.8 Causality, latency, and what a filter can know

Proposition 8.4. *A stable LTI filter with real impulse response is causal and has zero phase at every frequency only if it is a pure gain, $h[n] = c\,\delta[n]$.*

Proof. Zero phase with real coefficients means $H(e^{i\omega})$ is real, which for a real impulse response is equivalent to $h[n] = h[-n]$. Causality requires $h[n] = 0$ for $n < 0$, hence also for $n > 0$. □

Every real-time filter therefore introduces delay or phase distortion. Closed-loop experiments that trigger stimulation on an oscillatory phase must estimate the phase from past samples only, typically by autoregressive forward prediction or by causal filters with known group delay, and should report the achieved phase accuracy measured offline with a zero-phase filter on the same data.

Example 8.5 (Group delay of a theta filter)**.** A symmetric FIR band-pass for 6–10 Hz at $f_s = 1000$ Hz with $L = 1001$ taps has group delay 500 ms, four theta cycles. A causal implementation can only report the theta phase of half a second ago. An IIR Butterworth of order 2 has much shorter but frequency-dependent group delay, of the order of tens of milliseconds near the passband centre and larger at the edges, so phase estimates are biased differently for 6 Hz and 10 Hz components.

Exercises for this chapter

Exercise 8.6. Show that an antisymmetric FIR filter, $h[k] = -h[L-1-k]$, also has linear phase, with an extra constant phase of $\pi/2$.

Exercise 8.7. Find the pre-warped analog cutoff for a digital low-pass at 100 Hz with $f_s = 1000$ Hz.

Exercise 8.8. For a notch with $r = 0.99$ at $f_s = 1000$ Hz, estimate the notch bandwidth and the ringing duration.

Exercise 8.9. Show that the frequency response of a forward–backward filter is $|H(e^{i\omega})|^2$.

Part IV

Spectra, spike trains, and neural dynamics

Chapter 9

Harmonic analysis and spectral time series

A continuous signal is a function $x : \mathbb{R} \to \mathbb{R}$ or $\mathbb{C}$. The Fourier transform, when meaningful in the appropriate sense, is $X(f) = \int_{\mathbb{R}} x(t)e^{-2\pi ift}\,dt$. A length-$N$ recording is a different object: a discrete sequence. An FFT of the recorded sequence does not directly reveal a unique continuous generating signal without sampling assumptions.

Result 57: Euler's formula and Fourier orthogonality. *Precise setting:* $e^{i\theta} = \cos\theta + i\sin\theta$ and $\int_0^T e^{2\pi i(k-\ell)t/T}\,dt = 0$ for distinct integers k, ℓ.

Why it matters here: Derive discrete spectral coefficients and phase representations.

Result 58: Parseval's identity and Plancherel's theorem. *Precise setting:* For appropriate Fourier expansions/transforms, energy in time equals energy in frequency, e.g. $\|x\|_2^2 = \|X\|_2^2$ under a unitary convention.

Why it matters here: Normalize power and check spectral energy.

Result 59: Fourier convolution theorem. *Precise setting:* For suitable integrable functions, $\mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g)$, with conventions fixed.

Why it matters here: Relate a digital filter's impulse response to its transfer function.

Result 60: Fourier shift and modulation identities. *Precise setting:* Translation of a signal multiplies its spectrum by a complex phase; multiplication by a complex sinusoid translates frequency.

Why it matters here: Understand lags, phase offsets, and analytic demodulation.

Result 61: Heisenberg–Gabor uncertainty inequality. *Precise setting:* With standard centering and Fourier conventions, a nonzero finite-energy waveform has time spread and frequency spread whose product obeys a nonzero lower bound.

Why it matters here: Understand the temporal and spectral resolution trade-off.

Result 62: Paley–Wiener theorems. *Precise setting:* Appropriate compact-support and exponential-type conditions characterize special classes of Fourier transforms; compactly supported time-domain signals have entire Fourier transforms of controlled exponential growth under standard hypotheses. In particular, nonzero ordinary signals cannot have both time and frequency support compact: the identity theorem supplies this corollary.

Why it matters here: Explain the role of analyticity in exact localization; Benedicks-type results strengthen restrictions for finite-measure supports.

Result 63: Riemann–Lebesgue lemma. *Precise setting:* The Fourier transform of an L^1 function vanishes at infinity.

Why it matters here: Understand high-frequency decay under integrability assumptions.

Result 64: Bochner and Herglotz spectral representation. *Precise setting:* Continuous positive-definite functions on $\mathbb{R}$ are Fourier transforms of finite nonnegative measures; discrete-time analogues use nonnegative measures on the frequency circle.

Why it matters here: Characterize admissible stationary covariance functions and spectra.

Result 65: Cramer's spectral representation. *Precise setting:* A zero-mean weakly stationary second-order process can be represented via orthogonal-increment random spectral measures under the standard hypotheses.

Why it matters here: Interpret stochastic LFP spectra rather than deterministic FFTs alone.

Result 66: Wold decomposition. *Precise setting:* A purely nondeterministic covariance-stationary discrete-time process has a one-sided linear innovations representation; general stationary processes decompose into deterministic and purely nondeterministic parts.

Why it matters here: Understand assumptions behind linear innovations and VAR models.

Result 67: Yule–Walker equations. *Precise setting:* For a stationary autoregressive process, lagged covariance satisfies linear equations determined by AR coefficients.

Why it matters here: Relate autocovariance to AR spectral estimators.

Result 68: Wiener–Hopf normal equations. *Precise setting:* The least-squares optimal linear predictor has a residual orthogonal to the predictor space, yielding covariance normal equations.

Why it matters here: Derive linear decoding and linear prediction filters.

Result 69: Spectral coherence bound. *Precise setting:* A positive-semidefinite two-signal spectral matrix satisfies $|S_{xy}(f)|^2 \le S_{xx}(f)S_{yy}(f)$ where the spectra exist; hence $0 \le C_{xy}(f) \le 1$ for positive denominators.

Why it matters here: Interpret magnitude-squared LFP coherence.

Result 70: Hilbert-transform Fourier multiplier identity. *Precise setting:* On $L^2(\mathbb{R})$, with appropriate Fourier convention, $\mathcal{F}(\mathcal{H}x)(f) = -i\,\mathrm{sgn}(f)\mathcal{F}x(f)$.

Why it matters here: Build analytic signals and interpret instantaneous phase.

Definition 9.1. For zero-mean jointly wide-sense stationary signals x, y, let $R_{xy}(\tau) = \mathbb{E}[x(t+\tau)\overline{y(t)}]$. Under appropriate integrability, $S_{xy} = \mathcal{F}R_{xy}$. Magnitude-squared coherence is $C_{xy} = |S_{xy}|^2/(S_{xx}S_{yy})$ wherever the denominator is strictly positive.

Interpretation warning 9.2. Coherence is spectral dependence, not anatomical connectivity. Common reference, volume conduction, leakage, sample-size bias, and nonstationarity can distort interpretation [4, 8].

9.1 Proofs of selected harmonic-analysis results

Proof of Result 59 for integrable functions. For $f, g \in L^1(\mathbb{R})$, Tonelli's theorem gives

$$\iint |f(s)||g(t-s)|\,ds\,dt = \|f\|_1 \|g\|_1 < \infty,$$

so $f * g \in L^1$ and Fubini allows the exchange

$$\mathcal{F}(f*g)(\xi) = \iint f(s)g(t-s)e^{-2\pi i\xi t}\,ds\,dt = \int f(s)e^{-2\pi i\xi s}\Big(\int g(u)e^{-2\pi i\xi u}\,du\Big)ds,$$

after the substitution $u = t - s$. The inner integral is $\mathcal{F}g(\xi)$. □

Proof of Result 63 (Riemann–Lebesgue). For $f = \mathbf{1}_{[a,b]}$, $|\mathcal{F}f(\xi)| = |e^{-2\pi i\xi a} - e^{-2\pi i\xi b}|/(2\pi|\xi|) \le 1/(\pi|\xi|) \to 0$. By linearity the claim holds for step functions. Every $f \in L^1$ is an L^1-limit of step functions f_n, and $\sup_\xi |\mathcal{F}f(\xi) - \mathcal{F}f_n(\xi)| \le \|f - f_n\|_1$, so the limit property passes to f. □

The rate of decay reflects smoothness: a jump gives $1/|\xi|$ decay, a kink $1/|\xi|^2$, and a function with k integrable derivatives decays faster than $|\xi|^{-k}$. Sharp spike waveforms and stimulation artefacts therefore spread power over a broad high-frequency range, which is why they contaminate gamma and high-frequency band estimates.

9.2 Autoregressive spectra and Yule–Walker

For a stationary AR(p) process $X_t = \sum_{k=1}^{p} a_k X_{t-k} + \varepsilon_t$ with white innovations of variance σ^2, multiplying by X_{t-m} ($m \ge 1$) and taking expectations gives Result 67:

$$\gamma(m) = \sum_{k=1}^{p} a_k \gamma(m-k), \qquad m = 1, \dots, p,$$

and $\gamma(0) = \sum_k a_k \gamma(k) + \sigma^2$. Solving these Toeplitz equations estimates the coefficients from sample autocovariances. The spectral density is

$$S(f) = \frac{\sigma^2/f_s}{\big|1 - \sum_{k=1}^{p} a_k e^{-2\pi i k f/f_s}\big|^2}.$$

Example 9.3 (AR(2) resonance)**.** For $X_t = 2r\cos\theta\, X_{t-1} - r^2 X_{t-2} + \varepsilon_t$ with $0 < r < 1$, the characteristic polynomial has roots $re^{\pm i\theta}$. The spectrum has a peak near $f_0 = \theta f_s/(2\pi)$ whose width is approximately $(1-r)f_s/\pi$, and the autocorrelation is a damped cosine $\propto r^{|k|}\cos(\theta|k| + \psi)$. With $f_s = 250\,\text{Hz}$, θ chosen for $8\,\text{Hz}$ and $r = 0.98$, the process looks like a waxing and waning theta rhythm with a $1.6\,\text{Hz}$-wide spectral peak, yet it is a linear stochastic resonance: there is no limit cycle, and amplitude and phase are driven entirely by noise. AR(2) processes are the standard null model for "is this oscillation more than filtered noise?".

9.3 Wiener–Hopf prediction and decoding

The best linear predictor of Y_t from X_{t-k}, $k = 0, \dots, K$, is $\hat{Y}_t = \sum_k w_k X_{t-k}$ with weights solving the normal equations (Result 68)

$$\sum_{k=0}^{K} w_k\, \gamma_{xx}(j-k) = \gamma_{yx}(j), \qquad j = 0, \dots, K.$$

Proof. The squared error is a convex quadratic in w. Setting its derivative with respect to w_j to zero is equivalent to $\mathbb{E}[(Y_t - \hat{Y}_t)X_{t-j}] = 0$: the residual is orthogonal to every regressor, which is the orthogonal projection theorem (Result 26). □

With unlimited lags the solution in frequency is the Wiener filter $W(f) = S_{yx}(f)/S_{xx}(f)$ (non-causal) and the explained fraction of variance at frequency f is the coherence $C_{yx}(f)$. Coherence is therefore the fraction of Y's power at f that is linearly predictable from X *at all lags, past and future*: it is a symmetric measure of linear dependence, not of directed influence.

9.4 Shift, modulation, and the meaning of phase

By Result 60, $x(t-\tau)$ has transform $e^{-2\pi i f\tau}X(f)$. A pure delay shows up as a phase that is linear in frequency with slope $-2\pi\tau$. Conversely, if the cross-spectrum phase between two channels is approximately linear in f over a band, its slope gives a delay estimate (the *phase-slope index* is a robust version of this idea). A constant phase offset, independent of frequency, does not correspond to a time delay at all; it is what a reference subtraction, a derivative, or a Hilbert transform produces.

Exercises for this chapter

Exercise 9.4. Prove Parseval's identity for Fourier series on $[0, T]$ from the orthogonality relations of Result 57 for trigonometric polynomials.

Exercise 9.5. Derive the Yule–Walker equations for AR(1) and show that $\gamma(k) = a^{|k|}\sigma^2/(1-a^2)$.

Exercise 9.6. For the AR(2) example, find θ for an 8 Hz peak at $f_s = 250$ Hz and compute the coefficients.

Exercise 9.7. Show that $y(t) = x(t-\tau)$ has coherence one with x at every frequency and cross-spectral phase $-2\pi f\tau$.

Chapter 10

Time–frequency representations and estimator uncertainty

A discrete Fourier transform can be computed exactly while an estimate of an unknown stochastic power spectrum remains noisy. The distinction is central to electrophysiological inference.

10.1 Why the raw periodogram is not generally consistent

For a zero-mean Gaussian white-noise series with variance σ^2 and length T, define at a non-DC Fourier frequency

$$I_T(f_k) = \frac{1}{T}\left|\sum_{t=0}^{T-1} X_t e^{-2\pi i k t/T}\right|^2 .$$

At generic positive Fourier frequencies below Nyquist, real and imaginary Fourier coefficients are independent normals with the appropriate variance; hence $2I_T(f_k)/\sigma^2 \sim \chi^2_2$ under this exact white Gaussian model. Consequently the periodogram variance does not vanish as T increases. Averaging approximately independent tapered or segmented estimates reduces variance, at a cost of frequency resolution and possible bias. For K independent identical exponential spectral ordinates, $2K\bar{I}/S \sim \chi^2_{2K}$; Welch overlap and tapered spectra violate exact independence, so their effective degrees of freedom need calibration.

10.2 Short-time Fourier transform, wavelets and admissibility

A window w produces the short-time Fourier transform $V_w x(\tau, f) = \int x(t)\overline{w(t-\tau)}e^{-2\pi i f t}\,dt$. Window duration sets an uncertainty trade-off; the display resolution is not a ground-truth measurement of a neural oscillation's instantaneous frequency. For a continuous wavelet ψ, the usual admissibility condition is

$$C_\psi = \int_{\mathbb{R}} \frac{|\Psi(f)|^2}{|f|}\,df < \infty.$$

It requires zero mean for sufficiently regular wavelets, so an uncorrected finite-frequency Morlet wavelet is only approximately admissible; a DC correction makes the distinction explicit. Reconstruction follows with the appropriate wavelet normalization. Discrete wavelet frames require bounds $A\|x\|^2 \le \sum_j |\langle x, \psi_j\rangle|^2 \le B\|x\|^2$ for positive finite A, B.

10.3 Coherence, phase consistency and finite samples

Even when the population coherence is zero, the squared magnitude of a finite-sample coherence estimate is nonnegative and generally biased upward. A simple idealized average of K independent complex Gaussian spectral replicates gives, under zero population coherence, an estimator with $\mathbb{E}[\widehat{C}] \approx 1/K$; this is *not* a universal threshold for overlapped, tapered or strongly dependent data. For iid spike phases $\phi_1, \dots, \phi_N$, the usual phase-locking value (PLV) is $|N^{-1} \sum_j e^{i\phi_j}|$. Even uniformly distributed phases produce a positive finite-sample value. The pairwise phase consistency (PPC)

$$\widehat{\mathrm{PPC}} = \frac{1}{N(N-1)} \sum_{j \neq k} \cos(\phi_j - \phi_k)$$

has zero expectation for independent uniform phases, provided the sampling and independence conditions hold. Surrogates made by Fourier phase randomization typically preserve individual Fourier magnitudes; iterative amplitude-adjusted Fourier transforms (IAAFT) approximately preserve both the marginal amplitude distribution and a target power spectrum. Neither automatically preserves cross-channel dependence or experimental trial structure.

Example 10.1 (What is—and is not—implemented)**.** `welch_psd.m`, `multitaper_psd.m`, `wavelet_coherence.m` and `pairwise_phase_consistency.m` produce numerical estimators. A scientific significance statement requires effective degrees of freedom, trial/block design and an appropriate null; it is not guaranteed by the function name.

10.4 Morlet wavelets: resolution and normalization

A complex Morlet wavelet centred at frequency f_0 with n_c cycles is

$$\psi(t) = A \exp(2\pi i f_0 t) \exp\!\left(-t^2/(2\sigma_t^2)\right), \qquad \sigma_t = \frac{n_c}{2\pi f_0}.$$

Its Fourier transform is a Gaussian centred at f_0 with standard deviation $\sigma_f = 1/(2\pi\sigma_t) = f_0/n_c$, so that $\sigma_t \sigma_f = 1/(2\pi)$. The energy densities $|\psi|^2$ and $|\Psi|^2$ have standard deviations $\sigma_t/\sqrt{2}$ and $\sigma_f/\sqrt{2}$, whose product $1/(4\pi)$ is the minimum allowed by the uncertainty principle (Result 61), as for every Gaussian. The *number of cycles* is the single parameter that sets the trade-off: at $f_0 = 40\,\mathrm{Hz}$ with $n_c = 7$, $\sigma_t \approx 28\,\mathrm{ms}$ and $\sigma_f \approx 5.7\,\mathrm{Hz}$; at $f_0 = 4\,\mathrm{Hz}$ the same n_c gives $\sigma_t \approx 280\,\mathrm{ms}$ and $\sigma_f \approx 0.57\,\mathrm{Hz}$. Constant n_c is constant relative bandwidth, the defining property of a wavelet transform.

Two normalizations are common. With $A = (\sigma_t\sqrt{\pi})^{-1/2}$ the wavelet has unit energy and white noise yields equal expected power at all scales. With A chosen so that $|\Psi(f_0)| = 1$, a unit-amplitude sinusoid at f_0 yields unit amplitude, which is convenient for envelope interpretation. Power spectra computed with the two conventions differ by a factor proportional to f_0, which changes the apparent $1/f$ slope by one. The convention must be reported.

10.5 Testing phase concentration

For N independent phases, the Rayleigh statistic $Z = NR^2$, with R the mean resultant length, is approximately exponential with mean one under uniformity; a refined approximation of the p-value is

$$p \approx \exp\!\Big(\sqrt{1 + 4N + 4(N^2 - NZ)} - (1 + 2N)\Big).$$

The test is most powerful against unimodal (von Mises) alternatives and has no power against bimodal locking with opposite modes, for which the resultant vanishes. The relation $\widehat{\mathrm{PPC}} = (NR^2 - 1)/(N - 1)$ connects the PPC of this chapter to the Rayleigh statistic, so the two measures carry the same information about a single set of phases; PPC is preferred for *comparing* conditions because its expectation does not depend on N.

Proposition 10.2. *For independent phases with common resultant $\rho = |\mathbb{E}e^{i\phi}|$, $\mathbb{E}[\widehat{\mathrm{PPC}}] = \rho^2$.*

Proof. For $j \neq k$, independence gives $\mathbb{E}e^{i(\phi_j - \phi_k)} = \mathbb{E}e^{i\phi_j}\overline{\mathbb{E}e^{i\phi_k}} = \rho^2$, which is real; its real part is $\mathbb{E}\cos(\phi_j - \phi_k)$. Average over the $N(N-1)$ ordered pairs. □

Spikes from the same burst, or many spikes from one trial, violate independence. The PPC variants that only pair spikes from *different trials* remove this dependence at the cost of variance.

10.6 Why phase-randomized surrogates preserve the spectrum

Given a real series of length T with DFT X_k, a phase-randomized surrogate replaces X_k by $|X_k|e^{i\theta_k}$ with independent uniform θ_k for $0 < k < T/2$, sets $\theta_{T-k} = -\theta_k$ to keep the result real, and inverts. Its periodogram is identical to the original's by construction, and its circular autocovariance is therefore identical too (the periodogram is the DFT of the circular sample autocovariance). Under the null hypothesis that the data are a linear Gaussian stationary process, the surrogates are draws from (approximately) the same distribution; nonlinear statistics such as PAC, bicoherence or time-asymmetry then have a valid reference distribution. The surrogate destroys every non-Gaussian feature, including spikes, sharp waves and bursts, so a rejection means only "not a stationary linear Gaussian process", which is almost always true of brain signals.

Exercises for this chapter

Exercise 10.3. Derive $\sigma_f = f_0/n_c$ for the Morlet wavelet from the Fourier transform of a Gaussian.

Exercise 10.4. Show that the mean resultant length of phases concentrated equally at 0 and π is zero, and propose a statistic that detects such bimodal locking.

Exercise 10.5. Prove that the periodogram is the DFT of the circular sample autocovariance.

Exercise 10.6. Simulate phases from a von Mises distribution with $\kappa = 0.5$ and compare the average PLV and PPC for $N = 10$ and $N = 1000$.

Chapter 11

Sampling, spectral concentration, and random-matrix theorems

The following classical theorems connect continuous-time measurement, spectral estimation, and high-dimensional statistical models. They are presented here immediately after Fourier and spectral foundations rather than as a revision-history appendix; each requires its precise hypotheses.

11.1 Nyquist–Shannon sampling

Theorem 11.1 (Classical bandlimited reconstruction)**.** *For a finite-energy continuous-time signal with Fourier support inside* $[-B, B]$ *Hz, an ideal uniform, noiseless sampling sequence at* $F_s > 2B$ *determines the signal. With* $\operatorname{sinc}(u) = \sin(\pi u)/(\pi u)$*, reconstruction has the form*

$$x(t) = \sum_{n \in \mathbb{Z}} x(n/F_s) \operatorname{sinc}(F_s t - n),$$

in the suitable convergence sense.

Application: set anti-aliasing and acquisition requirements. Real filters have finite rolloff; extracellular spikes are not literally compactly bandlimited.

11.2 Bedrosian's theorem

Theorem 11.2 (A sufficient spectral-separation case)**.** *If* a *has Fourier support in* $[-A, A]$ *and* c *is supported outside* $(-A, A)$*, with the required transform and product hypotheses, then* $\mathcal{H}(ac) = a\,\mathcal{H}(c)$*.*

For $c(t) = \cos(2\pi f_0 t)$ with $f_0 > A$, an ideal positive envelope $a(t)$ is recovered from the analytic signal of $a(t)c(t)$. The nonoverlap condition is sufficient, not a general necessary and sufficient criterion. PAC remains susceptible to waveform shape, transients, nonstationarity and filtering [1, 9].

11.3 Wiener–Khinchin theorem

Theorem 11.3. *A weakly stationary second-order process has an autocovariance and a corresponding nonnegative spectral measure. When densities exist and inversion is justified,*

$$S_x(f) = \int_{\mathbb{R}} R_x(\tau) e^{-2\pi i f \tau}\, d\tau, \qquad R_x(\tau) = \int_{\mathbb{R}} S_x(f) e^{2\pi i f \tau}\, df.$$

This supports power spectral estimation; it does not guarantee a unique oscillatory/aperiodic decomposition [10, 12].

11.4 Slepian concentration and DPSS tapers

For a length-N taper v and normalized half-bandwidth $0 < W < 1/2$, define the concentration ratio

$$\lambda(v) = \frac{\int_{-W}^{W} |\sum_{n=0}^{N-1} v_n e^{-2\pi i f n}|^2\, df}{\int_{-1/2}^{1/2} |\sum_{n=0}^{N-1} v_n e^{-2\pi i f n}|^2\, df}.$$

Theorem 11.4. *Eigenvectors of the discrete spectral-concentration operator maximize this ratio successively under orthogonality constraints. Its eigenvalues lie in* $[0, 1]$.

Application: `matlab/path/multitaper_psd.m` uses MATLAB DPSS and averages tapered periodograms, without adaptive taper weighting.

11.5 Johnson–Lindenstrauss lemma

Theorem 11.5 (Finite-set projection)**.** *A suitable normalized random linear map to dimension* $m = O(\varepsilon^{-2} \log P)$ *approximately preserves all pairwise Euclidean distances between* P *fixed points with high probability; constants also depend on the desired failure probability.*

A random dense projection is *not* equivalent to sampling a random subset of neural coordinates. Coordinate sampling needs additional coherence, leverage and sampling assumptions. This theorem does not alone certify reconstruction of neural manifolds.

11.6 Marcenko–Pastur and Tracy–Widom

Theorem 11.6 (Marcenko–Pastur law, standard isotropic model)**.** *For large sample covariance matrices from iid isotropic observations with ratio* $N/T \to \gamma$*, the continuous limiting spectrum is supported on* $[\sigma^2(1 - \sqrt{\gamma})^2, \sigma^2(1 + \sqrt{\gamma})^2]$*; when* $\gamma > 1$ *there is also an atom at zero.*

Theorem 11.7 (Tracy–Widom edge fluctuation)**.** *For suitable real Wishart ensembles, correctly centered and scaled largest-eigenvalue fluctuations at the upper edge follow a Tracy–Widom limit.*

Neither theorem provides an exact universal noise threshold for time-correlated, heteroscedastic, or preprocessed neural data. Edge fluctuations may occur on either side of the limiting edge.

11.7 Jensen's inequality

Theorem 11.8. *If ϕ is convex and the relevant expectations exist, $\phi(\mathbb{E}X) \le \mathbb{E}[\phi(X)]$.*

An elementary dimensionality bound follows instead directly from Cauchy–Schwarz:

$$1 \le D_{\mathrm{PR}} = \frac{(\sum_{i=1}^{r} \lambda_i)^2}{\sum_{i=1}^{r} \lambda_i^2} \le r, \quad \lambda_i > 0.$$

Jensen's more direct role in this textbook is the derivation of variational lower bounds and expectation-maximization, given in the GLM and optimization chapter. Inferring a structural biological dimensionality bound requires additional assumptions.

11.8 Schoenberg positive-definite radial kernels

Theorem 11.9 (Powered exponential, all Euclidean dimensions)**.** *For $\alpha > 0$, the kernel $k(x, y) = \exp(-\alpha\|x - y\|^q)$ is positive definite on all Euclidean dimensions for $0 < q \le 2$; this all-dimension guarantee does not hold for $q > 2$.*

This provides a covariance-valid kernel family for Gaussian processes, but not a proof that a particular model recovers physiological factors. Matern kernels are a different family with their own assumptions.

11.9 Tonelli and Fubini

Theorem 11.10 (Tonelli)**.** *For nonnegative measurable f over a product of sigma-finite measure spaces, iterated integrals can be interchanged, possibly with value $+\infty$.*

Theorem 11.11 (Fubini)**.** *For absolutely integrable measurable f on a product of sigma-finite measure spaces, the integrals can be interchanged and have finite common value.*

These theorems justify integrals used in spike-count expectations and latent-variable models only when their hypotheses apply.

11.10 The expected periodogram: Wiener–Khinchin in finite samples

Let X_t be zero-mean stationary with absolutely summable autocovariance γ and spectral density $S(f) = \sum_k \gamma(k) e^{-2\pi i f k}$ (unit sampling interval). For the periodogram $I_T(f) = T^{-1} |\sum_{t=0}^{T-1} X_t e^{-2\pi i f t}|^2$,

$$\mathbb{E} I_T(f) = \sum_{|k|<T} \Big(1 - \frac{|k|}{T}\Big) \gamma(k) e^{-2\pi i f k} = \int_{-1/2}^{1/2} F_T(f - \nu) S(\nu)\, d\nu,$$

where $F_T(f) = T^{-1} \big(\sin(\pi T f)/\sin(\pi f)\big)^2$ is the Fejér kernel.

Proof. Expand the squared modulus into a double sum over s, t and take expectations: $\mathbb{E} X_s X_t = \gamma(s - t)$. Grouping by $k = s - t$ there are $T - |k|$ pairs with each lag, giving the first expression. The second follows by substituting $\gamma(k) = \int S(\nu) e^{2\pi i \nu k} d\nu$ and recognising $\sum_{|k|<T} (1 - |k|/T) e^{2\pi i (\nu - f) k} = F_T(f - \nu)$. □

The periodogram is therefore an unbiased estimate not of $S(f)$ but of S smoothed by the Fejér kernel, whose sidelobes decay only as f^{-2}. Where the spectrum has a large dynamic range, such as the steep $1/f^{\chi}$ LFP spectrum, leakage from low frequencies can dominate the expected periodogram at high frequencies. Tapering replaces F_T by the squared transform of the taper, which is the rigorous reason that Slepian tapers reduce bias.

11.11 Johnson–Lindenstrauss in numbers

The lemma guarantees that n points can be mapped linearly to $k = O(\varepsilon^{-2}\log n)$ dimensions with all pairwise distances preserved within a factor $1 \pm \varepsilon$, and a random Gaussian matrix scaled by $1/\sqrt{k}$ achieves this with high probability. A commonly used explicit sufficient dimension is $k \geq 4\log n/(\varepsilon^2/2 - \varepsilon^3/3)$.

Example 11.12 (Random projections of population states)**.** For $n = 10^4$ time bins and $\varepsilon = 0.2$, the bound gives $k \geq 4 \cdot 9.21/(0.02 - 0.00267) \approx 2126$. The number does not depend on the number of recorded neurons. The lemma is therefore a statement that *distance geometry* of many points survives random compression, not that neural activity is low-dimensional: the required k grows with the number of points, whatever the data.

11.12 When Fubini fails

The hypotheses of Fubini's theorem are not decorative. On $[0,1]^2$, let $f(x,y) = (x^2 - y^2)/(x^2+y^2)^2$. Then $\int_0^1 \int_0^1 f\,dy\,dx = \pi/4$ while $\int_0^1 \int_0^1 f\,dx\,dy = -\pi/4$, because $\iint |f| = \infty$. In spike-train calculations the analogous failure arises when exchanging an expectation with an infinite sum over spikes whose contributions are not absolutely summable, for example in shot-noise models with heavy-tailed kernels.

Exercises for this chapter

Exercise 11.13. Show that $F_T(0) = T$ and $\int_{-1/2}^{1/2} F_T(f)\,df = 1$.

Exercise 11.14. For white noise, show that $\mathbb{E} I_T(f) = \gamma(0)$ exactly for every T.

Exercise 11.15. Compute the JL dimension for $n = 10^6$ and $\varepsilon = 0.1$.

Exercise 11.16. Verify the two iterated integrals in the Fubini counterexample.

Chapter 12

Counting processes, spike trains, and information

A spike train is a set of event times or, more generally, a random locally finite counting measure. Write $N((a, b])$ for the number of events in an interval. A conditional intensity $\lambda(t \mid \mathcal{H}_t)$ describes the conditional instantaneous event rate given the preceding history $\mathcal{H}_t$, if such a predictable intensity exists.

Result 71: Kolmogorov extension theorem. *Precise setting:* A consistent collection of finite-dimensional probability distributions defines a stochastic process on a suitable product measurable space.

Why it matters here: Build process models from their joint laws across finite time sets.

Result 72: Campbell's theorem. *Precise setting:* For a point process with mean measure Λ and nonnegative measurable h, $\mathbb{E}[\sum_{t_i} h(t_i)] = \int h \, d\Lambda$; signed integrable variants require absolute integrability.

Why it matters here: Compute expected spike-triggered sums and event statistics.

Result 73: Poisson superposition and thinning. *Precise setting:* Independent Poisson processes superpose to a Poisson process with the sum intensity; independent thinning with valid retention probability yields the corresponding Poisson intensity.

Why it matters here: Generate spike-train simulation benchmarks and control models.

Result 74: Doob–Meyer decomposition. *Precise setting:* A suitable integrable counting submartingale admits a predictable compensator $A(t)$, and $N(t) - A(t)$ is a martingale; when absolutely continuous, $A(t) = \int_0^t \lambda(s) \, ds$.

Why it matters here: Connect conditional intensity estimation and martingale residuals.

Result 75: Time-rescaling theorem. *Precise setting:* Under a correctly specified continuous conditional intensity and regularity of event times, transformed inter-event compensator increments are iid Exp(1) (with standard caveats for boundaries and conditioning).

Why it matters here: Construct goodness-of-fit diagnostics for spike-train GLMs.

Result 76: Conditional point-process likelihood. *Precise setting:* For a simple point process with conditional intensity λ on a finite observation interval, under standard domination assumptions its log-likelihood, up to reference terms, is $\sum_i \log \lambda(t_i) - \int \lambda(t) \, dt$.

Why it matters here: Derive maximum-likelihood fitting for spiking models.

Result 77: Palm theory and Slivnyak's theorem. *Precise setting:* For homogeneous Poisson processes, a reduced Palm process at a specified event has the law of the original process; more general Palm laws need not have this property.

Why it matters here: Understand event-centered descriptions and null spike correlations.

Result 78: Gibbs' inequality. *Precise setting:* For probability measures P, Q with $P \ll Q$, $D_{\mathrm{KL}}(P\|Q) \geq 0$, with equality if $P = Q$ almost everywhere.

Why it matters here: Understand KL-based divergence estimators and likelihood objectives.

Result 79: Data-processing inequality. *Precise setting:* For Markov chain $X \to Y \to Z$, $I(X;Z) \leq I(X;Y)$ whenever mutual information is defined.

Why it matters here: Characterize information lost through feature extraction or decoding.

Result 80: Fano's inequality. *Precise setting:* For a discrete class variable with $K \geq 2$ classes, entropy and classification error obey $H(X \mid \widehat{X}) \leq h_2(P_e) + P_e \log(K-1)$, with a consistent logarithm base.

Why it matters here: Relate decoder error to information; do not confuse with the spike-count Fano *factor*.

Example 12.1 (Count likelihood and binning)**.** With constant intensity λ on interval length Δ, $N_\Delta \sim \mathrm{Poisson}(\lambda\Delta)$. Then $\log p(n \mid \lambda) = n\log(\lambda\Delta) - \lambda\Delta - \log(n!)$. A binned GLM approximates continuous-time spiking only relative to its chosen bin width and assumptions about multiple spikes per bin.

Interpretation warning 12.2. A time-rescaling test diagnoses discrepancies from the hypothesized conditional intensity; a passed test is not proof of a unique physiological mechanism. Model misspecification can also arise from latent common drive or refractory dynamics.

12.1 Proofs of selected point-process results

Proof of Result 72 for simple functions and Poisson processes. Let N be a Poisson process with mean measure Λ and $h = \sum_j c_j \mathbf{1}_{B_j}$ with disjoint bounded sets B_j. Then $\sum_i h(t_i) = \sum_j c_j N(B_j)$ and $\mathbb{E}\sum_i h(t_i) = \sum_j c_j \Lambda(B_j) = \int h\, d\Lambda$. A nonnegative measurable h is an increasing limit of such simple functions, and monotone convergence (Result 22) extends the identity. The same proof works for any point process with mean measure Λ, since only $\mathbb{E}N(B) = \Lambda(B)$ was used. □

For Poisson processes the *variance* is equally explicit: $\mathrm{Var}\sum_i h(t_i) = \int h^2\, d\Lambda$. Applied to a spike-triggered sum $y(t) = \sum_i k(t - t_i)$ (shot noise), Campbell's theorem gives $\mathbb{E}y = \lambda \int k$ and $\mathrm{Var}\, y = \lambda \int k^2$ for rate λ: the mean grows with rate, and the coefficient of variation falls as $\lambda^{-1/2}$. This is Campbell's original application to vacuum-tube noise, and the basis of shot-noise LFP models.

Proof of Result 73 (superposition). If N_1, N_2 are independent Poisson processes with intensities λ_1, λ_2, then for disjoint sets the counts of $N = N_1 + N_2$ are sums of independent Poisson variables, hence Poisson with means $\Lambda_1(B) + \Lambda_2(B)$, and independent across disjoint

sets because each summand is. These are the defining properties of a Poisson process with intensity $\lambda_1 + \lambda_2$. □

Superposition explains why multi-unit activity summed over many weakly dependent neurons is often closer to Poisson than any single unit; the Palm–Khintchine theorem makes this precise for superpositions of many sparse independent renewal processes.

12.2 Deriving the point-process likelihood

Partition $[0, T]$ into bins of width Δ and approximate the conditional probability of a spike in bin k by $\lambda_k \Delta$, where $\lambda_k = \lambda(t_k \mid \mathcal{H}_{t_k})$. By the chain rule of conditional probability, the probability of the observed binary sequence (y_k) is

$$\prod_k (\lambda_k \Delta)^{y_k} (1 - \lambda_k \Delta)^{1 - y_k}.$$

Taking logarithms and using $\log(1 - \lambda_k \Delta) = -\lambda_k \Delta + O(\Delta^2)$,

$$\sum_k y_k \log \lambda_k + n \log \Delta - \sum_k \lambda_k \Delta + O(\Delta).$$

The term $n \log \Delta$ does not depend on the model. As $\Delta \to 0$, $\sum_k y_k \log \lambda_k \to \sum_i \log \lambda(t_i)$ and $\sum_k \lambda_k \Delta \to \int_0^T \lambda(t)\, dt$, which is Result 76. The derivation exposes the approximation made by a binned GLM: it is accurate when $\lambda_k \Delta \ll 1$ and when at most one spike falls in a bin. At 1 ms bins and 100 spikes/s, $\lambda \Delta = 0.1$ and the neglected terms are no longer negligible.

12.3 Information inequalities

Proof of Result 78 (Gibbs), discrete case. With $p_i > 0$ only where $q_i > 0$, Jensen's inequality for the concave logarithm gives $-D_{\mathrm{KL}}(p \| q) = \sum_i p_i \log(q_i / p_i) \le \log \sum_i p_i (q_i / p_i) \le \log 1 = 0$. Equality in Jensen requires q_i / p_i constant on the support of p, which together with normalization gives $p = q$. □

Proof of Result 79 (data processing). By the chain rule of mutual information, $I(X; Y, Z) = I(X; Z) + I(X; Y \mid Z) = I(X; Y) + I(X; Z \mid Y)$. For a Markov chain $X \to Y \to Z$, X and Z are conditionally independent given Y, so $I(X; Z \mid Y) = 0$. Hence $I(X; Z) = I(X; Y) - I(X; Y \mid Z) \le I(X; Y)$. □

Every stage of an analysis pipeline (sorting, binning, smoothing, dimensionality reduction) is a processing step $Y \to Z$ that cannot increase the information about the stimulus X. A decoder applied to binned counts cannot outperform an optimal decoder applied to the raw spike times. Information "gained" by a feature is information that a weaker decoder could not otherwise use, not information created.

Example 12.3 (Using Fano's inequality)**.** A decoder classifies 8 movement directions from population activity with error rate $P_e = 0.25$. Result 80 gives $H(X \mid \hat{X}) \le h_2(0.25) + 0.25 \log_2 7 \approx 0.811 + 0.702 = 1.51$ bits. With uniform directions $H(X) = 3$ bits, so $I(X; \hat{X}) \ge 3 - 1.51 = 1.49$ bits, and by data processing the population activity carries at least that much information about direction. Decoding accuracy thus gives a *lower* bound on information, not an estimate of it.

Exercises for this chapter

Exercise 12.4. Prove the Poisson case of Campbell's variance formula $\operatorname{Var} \sum_i h(t_i) = \int h^2 \, d\Lambda$ for simple functions.

Exercise 12.5. Compute the mean and variance of shot noise with exponential kernel $k(t) = Ae^{-t/\tau}\mathbf{1}_{t \geq 0}$ driven by a rate-λ Poisson process.

Exercise 12.6. Show that the binned Bernoulli log-likelihood and the binned Poisson log-likelihood differ by $O(\sum_k \lambda_k^2 \Delta^2)$.

Exercise 12.7. Prove that $I(X;Y) \geq 0$ from Gibbs' inequality.

Chapter 13

Renewal processes, spike-train spectra, and bin limits

A spike train may be treated as a random counting measure $n(t) = \sum_j \delta(t - t_j)$, where the Dirac deltas are *distributions*, not finite-amplitude voltage spikes. This representation connects point processes to Fourier spectra and to continuous-time LFP shot-noise models.

13.1 How Bernoulli bins approach a Poisson process

Partition a fixed duration T into $m = T/\Delta$ bins, each with independent event probability $p_\Delta = \lambda\Delta$. Then $N_T \sim \text{Binomial}(m, p_\Delta)$, with

$$\mathbb{P}(N_T = 0) = (1 - \lambda T/m)^m \longrightarrow e^{-\lambda T}.$$

The binomial probability generating function likewise converges to $\exp[\lambda T(s-1)]$, proving convergence of the *total count* to a Poisson distribution. Independence of disjoint bins and consistency across intervals give the homogeneous Poisson-process limit. A form of Le Cam's inequality bounds total-variation error in Poisson approximations by a constant multiple of $\sum_i p_i^2$; hence fixed T and $p_i = \lambda\Delta$ make this error vanish as $\Delta \to 0$. This approximation is not a guarantee for refractory or bursty neuronal processes.

13.2 Renewal variability

Suppose interspike intervals $S_1, S_2, \ldots$ are iid, strictly positive, with mean m and finite variance v. Their coefficient of variation is $CV^2 = v/m^2$. Under standard renewal limit conditions, for long observation windows T the spike count satisfies

$$\mathbb{E}[N(T)] \sim T/m, \qquad \text{Var}[N(T)] \sim (v/m^3)T, \qquad \frac{\text{Var}\, N(T)}{\mathbb{E}N(T)} \longrightarrow CV^2.$$

Thus *long-window* count Fano factor and ISI CV^2 are related for a renewal model. Nonrenewal history, trial mixing, latent gain, short windows and nonstationarity can break the empirical approximation.

13.3 Bartlett spectra and shot-noise LFP

For a stationary point process, the Bartlett spectrum is the spectrum of its centered event-counting measure, including the shot-noise contribution. A homogeneous Poisson train of rate λ has flat continuous spectral density $S_{nn}(f) = \lambda$ in a two-sided Hz convention (the DC mean is treated separately). For a stationary renewal process with inter-event characteristic function $\varphi(f) = \mathbb{E}[e^{-2\pi i f S}]$, the nonzero-frequency continuous spectral density takes the form

$$S_{nn}(f) = \lambda \frac{1 - |\varphi(f)|^2}{|1 - \varphi(f)|^2},$$

under the usual nonlattice and regularity assumptions. This shows how refractoriness or regular spiking produces spectral structure without sinusoidal membrane oscillations.

If an idealized LFP component is a linear superposition of spike-triggered kernels, $y(t) = \int h(t - s)\, dN(s)$, its centered spectrum satisfies $S_{yy}(f) = |H(f)|^2 S_{nn}(f)$; independent measurement noise adds its own spectrum. If both n and y derive from the same events, their cross-spectrum is proportional to $H(f)S_{nn}(f)$ with sign/conjugation depending on the cross-spectrum convention. This is an example of *spike–field coherence without a claim of synaptic causation.*

13.4 Hawkes excitation and stability

The classical branching formulation is described by Hawkes [14]. For a linear Hawkes intensity $\lambda(t) = \mu + \sum_{t_j < t} h(t - t_j)$ with $h \geq 0$, $\mu > 0$ and branching ratio $\nu = \int_0^\infty h(s)\, ds < 1$, a stationary version exists under the standard construction, and its mean rate is $\bar{\lambda} = \mu/(1 - \nu)$. When $\nu \geq 1$, this stationary finite-mean construction fails in the usual nonnegative linear model; the statement does not apply without alteration to inhibitory or nonlinear point-process intensities.

Example 13.1 (Arsenal interpretation)**.** Compare `spike_count_fano.m`, `spike_field_coherence.m` and `spike_field_ppc.m` on Poisson, renewal, and latent-rate simulations. A function operating on binned counts or spike phases does not itself implement a continuous-time Bartlett spectral estimator.

13.5 A central limit theorem for renewal counts

Theorem 13.2 (Renewal CLT)**.** *For a renewal process with interval mean m and variance $v < \infty$,*

$$\frac{N(T) - T/m}{\sqrt{vT/m^3}} \Longrightarrow \mathcal{N}(0, 1) \qquad (T \to \infty).$$

Proof sketch. Let $S_n = \tau_1 + \cdots + \tau_n$. The events $\{N(T) < n\}$ and $\{S_n > T\}$ coincide. Choose $n = n(T) = \lfloor T/m + x\sqrt{vT/m^3} \rfloor$. Then

$$\mathbb{P}\Big(\frac{N(T) - T/m}{\sqrt{vT/m^3}} < x\Big) \approx \mathbb{P}(S_n > T) = \mathbb{P}\Big(\frac{S_n - nm}{\sqrt{nv}} > \frac{T - nm}{\sqrt{nv}}\Big).$$

Since $T - nm \approx -xm\sqrt{vT/m^3}$ and $nv \approx vT/m$, the right-hand threshold tends to $-x$, and the ordinary CLT for S_n gives the limit $\mathbb{P}(Z > -x) = \Phi(x)$. Rounding errors in n vanish after scaling. □

The theorem justifies the long-window identity $\operatorname{Var} N(T) \sim vT/m^3$ used for the Fano factor. Its rate of convergence depends on the third moment of the intervals; for bursty neurons with heavy-tailed intervals it can be very slow, so the asymptotic Fano factor may not be reached in windows of a few seconds.

13.6 Rényi's characterization of the Poisson process

Theorem 13.3 (Rényi)**.** *A simple point process on $\mathbb{R}$ is a Poisson process with a diffuse mean measure Λ if and only if $\mathbb{P}(N(B) = 0) = e^{-\Lambda(B)}$ for every set B that is a finite union of intervals.*

The theorem says that the *void probabilities* alone determine a simple process to be Poisson: one need not check independence of counts or their full distributions. In practice it motivates a diagnostic: compute the fraction of windows of length w without spikes and compare it with $e^{-\hat{\lambda}w}$ across a range of w. For a stationary process with the same rate, $\mathbb{P}(N(w) = 0) = 1 - \bar{\lambda}w + \mathbb{P}(N(w) = 2) + 2\mathbb{P}(N(w) = 3) + \cdots$, so refractoriness and regular firing, which suppress multiple spikes in a window, *lower* the empty fraction relative to Poisson; bursting, which packs spikes together, raises it for windows longer than a burst.

13.7 Hawkes processes: mean rate and spectrum

For the linear Hawkes process of Section 13.4 with branching ratio $\nu < 1$, take expectations in the intensity: $\bar{\lambda} = \mu + \bar{\lambda}\int_0^\infty h(s)\,ds = \mu + \nu\bar{\lambda}$, which gives $\bar{\lambda} = \mu/(1-\nu)$. Each immigrant event starts a cluster of expected size $1/(1-\nu)$, matching the branching interpretation. The Bartlett spectrum is

$$S_{nn}(f) = \frac{\bar{\lambda}}{|1 - H(f)|^2}, \qquad H(f) = \int_0^\infty h(t)e^{-2\pi i f t}\,dt.$$

For an exponential kernel $h(t) = \nu\beta e^{-\beta t}$, $|1 - H(f)|^2$ is smallest at $f = 0$, so self-excitation concentrates power at low frequencies; at $f = 0$ the spectrum is $\bar{\lambda}/(1-\nu)^2$, giving a long-window Fano factor $1/(1-\nu)^2$. A branching ratio of 0.5 makes the Fano factor 4 even though each event triggers on average only half an extra event. Self-excitation is thus one mechanism of overdispersion, alongside latent gain fluctuations (Result 42); the two are hard to distinguish from counts alone, but differ in their autocorrelograms.

Example 13.4 (Gamma renewal spectrum)**.** For gamma intervals with shape κ and rate $\kappa\lambda$ (mean $1/\lambda$), $\varphi(f) = (1 + 2\pi i f/(\kappa\lambda))^{-\kappa}$. For $\kappa = 1$ (Poisson) the spectrum is flat. For large κ the train is nearly periodic and the spectrum develops peaks at multiples of the firing rate λ, with a dip at low frequencies: a regular neuron transfers little power below its own firing rate. A "spectral peak at the firing rate" in spike–field analysis may simply reflect regular firing, not an oscillatory input.

Exercises for this chapter

Exercise 13.5. Using the renewal CLT, compute an approximate 95% interval for the count of a gamma renewal neuron with rate 10 spikes/s and $\kappa = 4$ in 10 s.

Exercise 13.6. Verify Rényi's void probability for a homogeneous Poisson process and show that a dead-time process violates it for short windows.

Exercise 13.7. Compute $S_{nn}(f)$ for the exponential Hawkes kernel and find the frequency at which it falls to half its zero-frequency value.

Exercise 13.8. Show that for $\kappa = 1$ the gamma renewal spectrum formula gives $S_{nn}(f) = \lambda$.

Chapter 14

Exponential families, GLMs, and constrained optimization

The Poisson model in `poisson_spike_glm.m` can be derived from a probabilistic model, rather than treated as an unexplained numerical recipe.

14.1 Poisson counts as an exponential family

For count $Y \in \{0, 1, \ldots\}$ and positive mean μ,

$$p(y \mid \mu) = \exp(y \log \mu - \mu - \log(y!)).$$

Its natural parameter is $\eta = \log \mu$. For a row of covariates $x_i^\top$, exposure duration $\Delta_i > 0$, and regression coefficients β, define $\mu_i = \Delta_i \exp(x_i^\top \beta)$. In matrix form with design X,

$$\ell(\beta) = \sum_i [y_i(x_i^\top \beta + \log \Delta_i) - \Delta_i e^{x_i^\top \beta} - \log(y_i!)].$$

Proposition 14.1 (Concavity of the Poisson log likelihood)**.** *For fixed nonnegative counts and strictly positive exposures,*

$$\nabla \ell = X^\top (y - \mu), \qquad \nabla^2 \ell = -X^\top W X, \quad W = \operatorname{diag}(\mu_i) \succ 0.$$

Thus ℓ is concave; if X has full column rank, it is strictly concave at every finite coefficient vector.

Proof. Differentiate the exponential using

$$\frac{\partial \mu_i}{\partial \beta} = \mu_i x_i.$$

For nonzero v, $-v^\top \nabla^2 \ell v = \sum_i \mu_i (x_i^\top v)^2$, positive under full rank. □

Strict concavity is not existence of a finite maximizer. For all-zero counts and an unpenalized intercept, likelihood increases as the intercept tends to $-\infty$.

14.2 IRLS as Fisher scoring

Set $\eta = X\beta + \log \Delta$ and $\mu = e^{\eta}$. Newton scoring solves, at the current iterate,

$$(X^\top W X)\,\delta = X^\top (y - \mu), \qquad \beta_{\text{new}} = \beta + \delta.$$

For known exposure offset $o = \log \Delta$, the full working response is $z_{\text{full}} = \eta + (y - \mu)/\mu$. Weighted least squares on the design matrix X must use the offset-adjusted response:

$$z_{\text{WLS}} = z_{\text{full}} - o = X\beta + \frac{y - \mu}{\mu}, \qquad \beta_{\text{new}} = (X^\top W X)^{-1} X^\top W z_{\text{WLS}},$$

when the inverse exists. Alternatively fit z_{full} while holding o fixed as an offset, not as a fitted predictor. For the canonical Poisson family, the observed negative Hessian agrees with expected Fisher information. A quadratic ridge penalty adds $-\lambda P\beta$ to the score and λP to the negative Hessian. If the algorithm *clips* η, or lacks line search, the numerical iterations need not monotonically improve the original unclipped likelihood. This is relevant to the present IRLS draft.

14.3 Jensen, the ELBO, and EM monotonicity

For latent variable Z and any density $q(z)$ supported where the joint model is positive, Jensen gives

$$\log p_\theta(y) = \log \int q(z) \frac{p_\theta(y, z)}{q(z)}\, dz \geq \mathcal{L}(q, \theta) := \mathbb{E}_q[\log p_\theta(y, Z)] - \mathbb{E}_q[\log q(Z)].$$

The identity $\log p_\theta(y) = \mathcal{L}(q, \theta) + D_{\text{KL}}(q \| p_\theta(z \mid y))$ shows exactly when the lower bound is tight. Exact EM alternates an E-step setting q to the posterior and an M-step not decreasing the ELBO. This gives nondecreasing observed-data likelihood, but *not* a global optimum. For phase–amplitude coupling, the Tort modulation index is a normalized KL divergence between a binned phase-dependent amplitude distribution and the uniform distribution; positivity comes from Gibbs' inequality, while binning and surrogates are additional statistical decisions.

14.4 KKT and alternating least squares

For $\min_\beta f(\beta)$ with differentiable inequalities $g_j(\beta) \leq 0$ and affine equalities $A\beta = b$, the KKT system consists of primal feasibility, multipliers $\lambda_j \geq 0$, complementary slackness $\lambda_j g_j(\beta) = 0$, and stationarity

$$\nabla f(\beta) + \sum_j \lambda_j \nabla g_j(\beta) + A^\top \nu = 0.$$

Necessity for a local minimizer requires a suitable constraint qualification, such as LICQ. *Sufficiency* for global optimality holds when f and g_j are convex differentiable functions and the equality constraints are affine. Slater's condition is a standard qualification under its usual convex-program hypotheses. Lasso stationarity uses subgradients at zero. For a CP tensor decomposition, exact minimization over a factor block cannot increase the unpenalized objective *of that specified block problem*; alternating such steps does not make the multi-factor objective convex or guarantee global optimality. The exploratory `tensor_cp_als.m` includes ridge terms and does not track a monotonic objective; a monotonicity claim for that code must be tested, not assumed.

14.5 Deviance, model comparison, and overdispersion

The Poisson *deviance* compares a fitted model with the saturated model $\hat{\mu}_i = y_i$:

$$D = 2\sum_i \Big[y_i \log \frac{y_i}{\hat{\mu}_i} - (y_i - \hat{\mu}_i) \Big],$$

with the convention $0 \log 0 = 0$. For nested models the deviance difference is twice the log-likelihood ratio, approximately χ^2 with degrees of freedom equal to the difference in parameters (Result 52). Penalized comparisons use AIC $= -2\ell + 2p$ or BIC $= -2\ell + p \log n$; for spike trains, the effective n in BIC is ambiguous (bins? spikes? trials?), and cross-validated log-likelihood is usually more defensible.

If the data are overdispersed relative to Poisson, the MLE of β remains consistent when the mean model is correct, but its standard errors from $(X^\top W X)^{-1}$ are too small. A quasi-Poisson correction multiplies the covariance by the dispersion estimate $\hat{\phi} = \frac{1}{n-p}\sum_i (y_i - \hat{\mu}_i)^2/\hat{\mu}_i$; the sandwich estimator $(X^\top W X)^{-1}(X^\top \hat{\Omega} X)(X^\top W X)^{-1}$, with $\hat{\Omega} = \mathrm{diag}((y_i - \hat{\mu}_i)^2)$, is robust to any variance misspecification but not to temporal correlation of residuals, which calls for block-based variants.

14.6 Bernoulli GLMs and the logistic link

In fine bins where at most one spike can occur, $y_i \in \{0, 1\}$ and a Bernoulli model with $\mathbb{P}(y_i = 1) = \sigma(x_i^\top \beta)$, $\sigma(u) = 1/(1 + e^{-u})$, is natural. The log-likelihood $\sum_i [y_i x_i^\top \beta - \log(1 + e^{x_i^\top \beta})]$ is concave with Hessian $-X^\top \mathrm{diag}(p_i(1 - p_i))X$. For small probabilities $\sigma(u) \approx e^u$ and the model reduces to the Poisson GLM with log link. The *complementary log-log* link $\mathbb{P}(y_i = 1) = 1 - \exp(-\Delta e^{x_i^\top \beta})$ is exactly the probability of at least one event in a bin for a Poisson process with rate $e^{x_i^\top \beta}$, and is therefore the most faithful bin-level model of an underlying continuous-time intensity.

14.7 Sparse penalties and soft thresholding

For orthonormal design ($X^\top X = I$) and squared loss, the lasso problem $\min_\beta \frac{1}{2}\|y - X\beta\|^2 + \lambda\|\beta\|_1$ separates into scalar problems with solution

$$\hat{\beta}_j = \mathrm{sign}(z_j)\max(|z_j| - \lambda, 0), \qquad z = X^\top y.$$

Proof. For one coordinate minimize $\frac{1}{2}(\beta - z)^2 + \lambda|\beta|$. The subgradient condition $0 \in \beta - z + \lambda\,\partial|\beta|$ gives $\beta = z - \lambda$ if $z > \lambda$, $\beta = z + \lambda$ if $z < -\lambda$, and $\beta = 0$ when $|z| \le \lambda$, since then $0 \in -z + \lambda[-1, 1]$. □

For general designs, coordinate descent cycles this update with partial residuals; for GLMs it is applied inside each IRLS step. Lasso coupling filters are attractive because they return exact zeros, but which coupling coefficients are zero depends on correlations among regressors: among two strongly correlated presynaptic neurons, the lasso tends to select one arbitrarily. Zero coefficients are a property of the estimator, not evidence of absent connections.

14.8 An EM algorithm in closed form

Suppose each trial's count is drawn from one of two Poisson rates, $y_i \sim \pi\,\mathrm{Poisson}(\lambda_1) + (1-\pi)\,\mathrm{Poisson}(\lambda_2)$, for example from an up and a down state. With responsibilities

$$r_i = \frac{\pi\, p(y_i \mid \lambda_1)}{\pi\, p(y_i \mid \lambda_1) + (1-\pi)\, p(y_i \mid \lambda_2)}$$

(E-step), the M-step maximizes the expected complete-data log-likelihood and gives

$$\pi = \frac{1}{n}\sum_i r_i, \qquad \lambda_1 = \frac{\sum_i r_i y_i}{\sum_i r_i}, \qquad \lambda_2 = \frac{\sum_i (1-r_i) y_i}{\sum_i (1-r_i)}.$$

Each iteration does not decrease the observed-data likelihood (the ELBO argument of this chapter), but the likelihood has a symmetric copy under label swapping and may have degenerate maxima where one component collapses onto a single observation. Several initializations and a check that the fitted components are separated are required before interpreting them as "states".

Exercises for this chapter

Exercise 14.2. Show that the Poisson deviance is nonnegative and zero only for a perfect fit.

Exercise 14.3. Derive the Hessian of the Bernoulli log-likelihood and show it is negative semidefinite.

Exercise 14.4. Show that for small Δe^u the complementary log-log probability is approximately Δe^u.

Exercise 14.5. Derive the M-step formulas for the two-component Poisson mixture.

Chapter 15

Population geometry, detectability, and identifiability

A low-dimensional fit is not necessarily a detectable or uniquely identified neural mechanism. This chapter separates three questions: *Is a component present in the data? Can the estimator recover its direction? Does that direction have a unique biological meaning?*

15.1 Beyond the Marchenko–Pastur noise edge

Consider an idealized population covariance $\Sigma = I + \theta vv^\top$, with $\|v\| = 1$, $\theta > 0$, iid Gaussian observations, and $N/T \to \gamma \in (0, \infty)$. In the standard spiked-Wishart model, the top population eigenvalue is $1 + \theta$ and the sample-noise edge is $(1 + \sqrt{\gamma})^2$.

Theorem 15.1 (BBP transition, rank-one covariance case)**.** *When $\theta \le \sqrt{\gamma}$, the leading sample eigenvalue approaches the upper noise edge and the sample top direction does not have asymptotically nonzero alignment with v. When $\theta > \sqrt{\gamma}$, the sample leading eigenvalue converges to $(1 + \theta)(1 + \gamma/\theta)$, strictly above the edge, and asymptotic alignment is nonzero.*

The result belongs to the spiked-covariance phase-transition family [2]. It depends on the iid isotropic sampling model. Temporal dependence, heterogeneous neuron variances, preprocessing, and correlated noise modify its calibration. A detectable statistical spike is not automatically an identified cell assembly or an anatomical network.

15.2 Shrinkage is part of covariance estimation

For sample covariance S of N variables and a target T_0 (often proportional to the identity), a linear shrinkage estimator is $\widehat{\Sigma} = (1 - \alpha)S + \alpha T_0$. The optimal α depends on the loss, assumptions, and estimation method. Ledoit–Wolf-type procedures estimate shrinkage to reduce high-dimensional covariance risk [15]; James–Stein concerns shrinkage of normal means in its classical form. Neither is a license to tune hyperparameters on held-out test trials.

15.3 Nonuniqueness of latent coordinates

In Gaussian factor analysis, $y = \Lambda z + \epsilon$ with $z \sim \mathcal{N}(0, I)$, the rotation $\Lambda' = \Lambda Q$, $z' = Q^\top z$ for orthogonal Q leaves the observable covariance $\Lambda\Lambda^\top + \Psi$ unchanged. The latent axes are

therefore not uniquely labeled. For a linear state-space model,

$$x_{t+1} = Ax_t + Bu_t + w_t, \qquad y_t = Cx_t + v_t,$$

any invertible T changes coordinates to $A' = TAT^{-1}$, $B' = TB$, $C' = CT^{-1}$, with transformed process-noise covariance, while preserving the measured trajectories. This is *similarity ambiguity*, not an estimator bug. General nonlinear latent-variable models may have even larger coordinate freedom.

15.4 Alignment and comparison of subspaces

For matrices $X, Y \in \mathbb{R}^{n\times k}$ of aligned, centered samples, the orthogonal Procrustes problem is

$$\min_{Q^\top Q = I} \|XQ - Y\|_F^2.$$

Proposition 15.2 (Procrustes solution)**.** *Let $X^\top Y = U\Sigma V^\top$ be an SVD. Then $Q_* = UV^\top$ is an optimal orthogonal alignment.*

Proof. Expanding the squared norm leaves the variable term $-2\,\mathrm{tr}(Q^\top X^\top Y)$. Writing $Q = UZV^\top$, the trace is $\mathrm{tr}(Z^\top \Sigma) \le \sum_i \sigma_i$, maximized by $Z = I$. □

Canonical correlation analysis (CCA) instead finds directions with maximal *cross-view correlation* after whitening both covariance structures. Representational similarity analysis (RSA) and centered kernel alignment (CKA) compare geometries through different invariances. They require carefully matched observations, preprocessing and independent validation sets.

Example 15.3 (Existing code)**.** `population_pca.m` produces an SVD of centered recordings. `participation_ratio.m` summarizes a covariance eigenspectrum. Neither establishes BBP significance, latent-factor uniqueness, or a well-identified inter-area communication subspace without additional modeling and tests.

15.5 Canonical correlation analysis as an SVD

Let $X \in \mathbb{R}^{T\times p}$ and $Y \in \mathbb{R}^{T\times q}$ be centered, with covariance blocks $\Sigma_{XX}, \Sigma_{YY} \succ 0$ and Σ_{XY}. CCA seeks a, b maximizing $\mathrm{corr}(Xa, Yb)$.

Proposition 15.4. *The canonical correlations are the singular values of the whitened cross-covariance $K = \Sigma_{XX}^{-1/2}\Sigma_{XY}\Sigma_{YY}^{-1/2}$, and the canonical directions are $a = \Sigma_{XX}^{-1/2}u$, $b = \Sigma_{YY}^{-1/2}v$ for the corresponding singular vector pairs (u, v).*

Proof. Substitute $a = \Sigma_{XX}^{-1/2}u$ and $b = \Sigma_{YY}^{-1/2}v$. The correlation becomes $u^\top Kv/(\|u\|\|v\|)$, which by the variational characterization of singular values is maximized by the top singular pair with value σ_1. Subsequent pairs maximize subject to orthogonality, giving the remaining singular values. □

When p or q approaches T, the sample covariances are ill-conditioned and the leading sample canonical correlations approach one even for independent data. Regularized CCA replaces Σ_{XX} by $\Sigma_{XX} + \kappa I$, and canonical correlations must be assessed on held-out data. Two brain areas with "canonical correlation 0.9" in-sample may share almost nothing out of sample.

15.6 Reduced-rank regression

For $Y \approx XB$ with $\operatorname{rank} B \le r$, minimizing $\|Y - XB\|_F^2$ has a closed form. Let $\hat{B}_{\mathrm{OLS}} = (X^\top X)^{-1} X^\top Y$ and let V_r hold the top r right singular vectors of the fitted values $X\hat{B}_{\mathrm{OLS}}$. Then $\hat{B}_r = \hat{B}_{\mathrm{OLS}} V_r V_r^\top$.

Proof sketch. $\|Y - XB\|_F^2 = \|Y - X\hat{B}_{\mathrm{OLS}}\|_F^2 + \|X\hat{B}_{\mathrm{OLS}} - XB\|_F^2$ by orthogonality of the OLS residual to $\mathrm{col}(X)$. The second term is minimized over rank-r matrices XB by Eckart–Young (Result 29) applied to $X\hat{B}_{\mathrm{OLS}}$; the minimizer is its projection onto V_r, which is $X\hat{B}_{\mathrm{OLS}} V_r V_r^\top$. □

The row space of $\hat{B}_r$ (spanned by V_r) is the *predictive subspace* of the target area. Its dimension, chosen by cross-validation, is the "communication subspace" dimension in inter-areal studies. It measures how many linear combinations of the target are predictable from the source; it does not say that the source causes those combinations.

15.7 Centered kernel alignment and RSA

For representations $X \in \mathbb{R}^{n \times p}$, $Y \in \mathbb{R}^{n \times q}$ of the same n stimuli (columns centered), linear CKA is

$$\mathrm{CKA}(X, Y) = \frac{\|Y^\top X\|_F^2}{\|X^\top X\|_F \, \|Y^\top Y\|_F}.$$

It is invariant to orthogonal transformations and isotropic scaling of either representation, but not to arbitrary invertible linear maps (unlike CCA, which is invariant to them). This intermediate invariance is often what one wants when comparing neural populations whose individual neurons have no correspondence but whose relative variance across directions is meaningful. RSA compares the $n \times n$ dissimilarity matrices of the two representations with a rank correlation; it is invariant to any transformation preserving the chosen dissimilarity ordering. Both methods require the same stimuli in the same order, and both are biased by noise in the dissimilarity estimates: cross-validated (split-half) distances remove the positive bias that noise adds to every Euclidean distance.

Example 15.5 (A spiked model below and above threshold)**.** With $N = 200$ neurons, $T = 400$ samples ($\gamma = 0.5$) and a planted rank-one spike of strength θ, the BBP threshold is $\sqrt{0.5} \approx 0.71$. A spike with $\theta = 0.5$ (variance 1.5 in its direction, versus 1 elsewhere) is undetectable by the top eigenvalue: that eigenvalue sits near the noise edge $(1+\sqrt{0.5})^2 \approx 2.91$. With $\theta = 2$, the top eigenvalue converges to $(1 + 2)(1 + 0.25) = 3.75$, clearly above the edge, and the squared alignment of the sample eigenvector with the planted direction is $(1 - \gamma/\theta^2)/(1 + \gamma/\theta) = 0.875/1.25 = 0.7$. Detectability and direction recovery both depend on θ *relative to* $\sqrt{\gamma}$, so doubling the number of samples can turn an invisible component into a reliable one.

Exercises for this chapter

Exercise 15.6. Show that canonical correlations are invariant to invertible linear transformations of X and of Y.

Exercise 15.7. Show that linear CKA is invariant to $X \mapsto cXQ$ for orthogonal Q and $c \neq 0$, and give a linear map that changes it.

Exercise 15.8. In the spiked example, find the smallest T for which a spike of strength $\theta = 0.5$ becomes detectable with $N = 200$.

Exercise 15.9. Derive the Procrustes solution when reflections are not allowed ($\det Q = 1$).

Chapter 16

Dependent observations, selective inference, and causality

Time bins from one animal are typically correlated. Repeating a test at many channels, lags and frequency bands also creates selection effects. A valid mathematical result must state which dependence and selection operations it covers.

16.1 Variance of a time-series average

For a zero-mean weakly stationary time series with $\gamma(k) = \mathrm{Cov}(X_t, X_{t+k})$, direct expansion gives

$$\mathrm{Var}(\bar{X}_T) = \frac{1}{T}\left[\gamma(0) + 2\sum_{k=1}^{T-1}\left(1 - \frac{k}{T}\right)\gamma(k)\right].$$

This *finite-sample identity* needs no central limit theorem. When $\sum_k |\gamma(k)| < \infty$, the asymptotic variance is $[\gamma(0) + 2\sum_{k\geq 1}\gamma(k)]/T$. For normalized autocorrelation $\rho(k)$, an approximate effective sample size is $T_{\mathrm{eff}} = T/[1 + 2\sum_{k\geq 1}\rho(k)]$, provided that series is meaningful. Negative correlations can make this heuristic exceed T; it must not be blindly clipped or treated as an exact count of independent animals.

16.2 HAC and resampling under dependence

A heteroskedasticity-and-autocorrelation-consistent (HAC) variance estimator uses weighted autocovariances, for example

$$\widehat{\Omega} = \widehat{\gamma}(0) + 2\sum_{k=1}^{L}\left(1 - \frac{k}{L+1}\right)\widehat{\gamma}(k).$$

This is a Bartlett-window Newey–West form in a scalar, mean-estimation setting. Consistency requires hypotheses on stationarity, weak dependence, moments and bandwidth growth; finite samples and nonstationary trials require scrutiny. Block bootstrap preserves within-block temporal structure but is only justified under suitable dependence assumptions and a suitable block length. Randomly permuting every LFP sample is generally *not* an exchangeable null.

16.3 Multiple comparisons and post-selection

Holm's step-down procedure controls family-wise error under valid marginal p-values without an independence assumption. Benjamini–Yekutieli controls FDR under arbitrary dependence using the harmonic correction $c_m = \sum_{j=1}^{m} 1/j$ in its standard formulation; it may be conservative. A maximum-statistic permutation test controls a family of tests if the randomization acts on the units exchangeable under the joint null. Cluster-permutation procedures use adjacency and a cluster-level statistic; rejection does not locate the exact significant onset, offset, or single electrode. After selecting the largest spectral peak and testing it on the same data, ordinary unadjusted p-values generally fail to reflect that selection. Sample splitting, independent replication or mathematically justified selective inference are possible remedies.

16.4 Observational dependence versus intervention

A directed acyclic graph (DAG) encodes a collection of conditional-independence relations under the Markov property. Paths are blocked or unblocked according to *d-separation*. If Z is an appropriate backdoor adjustment set for a proposed causal effect $X \to Y$, under causal sufficiency assumptions relative to the DAG and positivity,

$$p(y \mid \mathrm{do}(x)) = \int p(y \mid x, z) p(z)\, dz.$$

The formula fails for invalid adjustment sets, unmeasured confounding, or violated structural assumptions. In particular, a lagged correlation does not supply the DAG. Faithfulness is an additional assumption (and can fail by parameter cancellation); the common-cause principle motivates but does not uniquely identify latent confounders.

16.5 Transfer entropy and Gaussian Granger

Transfer entropy from X to Y, conditional on selected histories Y^- and Z^-, is a conditional mutual information: $TE = I(Y_t; X^- \mid Y^-, Z^-)$. For jointly Gaussian variables, with unrestricted correct linear conditional predictors and covariance determinants specified consistently,

$$TE_{X \to Y \mid Z} = \tfrac{1}{2} \log \frac{\det \Sigma_{\text{reduced}}}{\det \Sigma_{\text{full}}} \quad \text{(nats)}.$$

This Gaussian equivalence is established in [3]. A common multivariate Granger measure is the full log ratio; therefore it equals $2TE$ in this Gaussian setting. This is a theorem about *predictive distributions*, not an identification theorem for causal interventions. Spectral Granger additionally requires consistent VAR transfer functions or spectral factorization (e.g., Wilson methods) and model residual checks. The present `granger_causality.m` computes a temporal log-variance ratio, not spectral Granger; the existing `transfer_entropy.m` is a first-order, discrete plug-in conditional-information estimator.

Interpretation warning 16.1. The two MATLAB primitives are exploratory and do not by themselves establish causal identifiability.

16.6 Holm's procedure controls the family-wise error

Proposition 16.2. *Let $p_{(1)} \le \cdots \le p_{(m)}$ be ordered p-values with hypotheses $H_{(i)}$. Reject $H_{(1)}, \ldots, H_{(k-1)}$ where k is the first index with $p_{(k)} > \alpha/(m-k+1)$. If every null p-value is*

valid ($\mathbb{P}(p \le u) \le u$), then $\mathbb{P}$(any true null rejected) $\le \alpha$, whatever the dependence.

Proof. Let I_0 be the set of m_0 true nulls and E the event that every p_i with $i \in I_0$ exceeds α/m_0. By the union bound, $\mathbb{P}(E^c) \le m_0 \cdot \alpha/m_0 = \alpha$. On E, consider the first true null in the ordered list, at position j. At most $m - m_0$ hypotheses precede it, so $j \le m - m_0 + 1$ and its threshold is $\alpha/(m - j + 1) \le \alpha/m_0$, which its p-value exceeds. The procedure stops there or earlier and rejects no true null. □

16.7 d-separation by example

Three elementary configurations determine whether a path between X and Y transmits association:

- a *chain* $X \to M \to Y$ and a *fork* $X \leftarrow C \to Y$ are open unless the middle node is conditioned on;
- a *collider* $X \to K \leftarrow Y$ is closed unless K or one of its descendants is conditioned on.

A path is blocked by a conditioning set Z if it contains a non-collider in Z or a collider with neither it nor any descendant in Z.

Example 16.3 (Conditioning on a detected event)**.** Let X and Y be the activities of two independent neurons, and let a "population event" K be detected when the summed activity crosses a threshold. Analysing only event-triggered epochs conditions on K, a collider: within events, the two neurons become negatively associated (if one is quiet, the other must have been active to trigger the event). Event-triggered correlations therefore cannot be compared with unconditional ones without modelling the selection.

Example 16.4 (Behaviour as a common cause)**.** If running speed C modulates both an LFP band X and a unit's rate Y, the fork $X \leftarrow C \to Y$ produces spike–field correlations. Adjusting for C (the backdoor formula of this chapter) blocks the path. If speed is measured noisily, adjustment is incomplete, and residual association persists: measurement error in confounders is a common reason why "controlling for behaviour" fails.

16.8 Spectral factorization and Wilson's method

For a bivariate stationary process with spectral density matrix $S(f)$, Wold's theorem (Result 66) and its spectral counterpart give a *minimum-phase factorization*

$$S(f) = H(f)\Sigma H(f)^*,$$

with $H(f) = I + \sum_{k \ge 1} H_k e^{-2\pi i f k}$ analytic in the sense that its inverse exists as a causal filter, and Σ the innovation covariance. For a scalar spectrum the factor has an explicit expression, $\log \sigma^2 = \int_{-1/2}^{1/2} \log S(f)\, df$ (the Kolmogorov–Szegő formula), which gives the one-step prediction error variance from the spectrum alone. In the multivariate case *Wilson's algorithm* computes H and Σ by Newton iterations on the equation $S = H\Sigma H^*$ on a frequency grid, preserving the minimum-phase property at each step. The resulting H and Σ feed Geweke's spectral Granger measure without fitting a finite VAR, avoiding the choice of model order but inheriting the bias and variance of the nonparametric spectral estimate, including its frequency smoothing.

Proposition 16.5 (Kolmogorov–Szegő for AR(1))**.** *For $X_t = aX_{t-1} + \varepsilon_t$ with $|a| < 1$ and innovation variance σ^2, $\int_{-1/2}^{1/2} \log S(f)\, df = \log \sigma^2$.*

Proof. $S(f) = \sigma^2 / |1 - a e^{-2\pi i f}|^2$, so the integral is $\log \sigma^2 - 2\Re \int \log(1 - a e^{-2\pi i f})\, df$. Expanding $\log(1 - az) = -\sum_{k \geq 1} a^k z^k / k$ and integrating term by term over the unit circle, every term vanishes. □

Exercises for this chapter

Exercise 16.6. Apply Holm and Bonferroni to five p-values of your choice and find a case where Holm rejects more hypotheses.

Exercise 16.7. For the DAG $C \to X \to Y$, $C \to Y$, show that $\{C\}$ is a valid backdoor adjustment set and that $\varnothing$ is not.

Exercise 16.8. Simulate two independent Gaussian sequences, select samples where their sum exceeds 2, and compute their correlation within the selection.

Exercise 16.9. Compute the effective sample size of an AR(1) series with $a = 0.8$ and $T = 500$.

Chapter 17

Dynamical systems, inverse problems, and neural manifolds

A state vector $x(t)$ represents a point in a state space. An ordinary differential equation (ODE) $\dot{x} = f(x, t)$ defines how that point moves, provided a solution exists. A vector of measured rates or LFP values is an observation of the state, not necessarily the complete physical state itself.

Result 81: Picard–Lindelof existence and uniqueness. *Precise setting:* If $f(t, x)$ is locally Lipschitz in x and continuous in t on a suitable neighborhood, a unique local solution exists through a given initial condition.

Why it matters here: Determine when deterministic neural state equations define a well-posed trajectory.

Result 82: Gronwall's inequality. *Precise setting:* A nonnegative function bounded by an integral inequality has an exponential comparison bound under its hypotheses.

Why it matters here: Quantify growth of perturbations to trajectories and models.

Result 83: Lyapunov stability theorems. *Precise setting:* A differentiable positive-definite function with suitable nonpositive or negative orbital derivative implies stability or asymptotic stability under the corresponding assumptions.

Why it matters here: Assess recurrent-circuit attractors and locally stable activity.

Result 84: LaSalle's invariance principle. *Precise setting:* In a compact positively invariant region with $\dot{V} \leq 0$, trajectories approach the largest invariant set where $\dot{V} = 0$, subject to standard hypotheses.

Why it matters here: Study convergence when strict energy decrease is unavailable.

Result 85: Hartman–Grobman theorem. *Precise setting:* Near a hyperbolic equilibrium, a smooth autonomous system is locally topologically conjugate to its linearization.

Why it matters here: Interpret local fixed-point geometry while respecting nonlinear effects.

Result 86: Kalman observability rank test. *Precise setting:* The linear time-invariant pair (A, C) with n states is observable exactly when $\operatorname{rank}[C^\top, (CA)^\top, \ldots, (CA^{n-1})^\top]^\top = n$.

Why it matters here: Assess recoverability of latent states from neural measurements.

Result 87: Kalman controllability rank test. *Precise setting:* The linear time-invariant pair (A, B) is controllable exactly when $\operatorname{rank}[B, AB, \ldots, A^{n-1}B] = n$.

Why it matters here: Characterize reachable states under modeled input interventions.

Result 88: Kalman filter optimality for linear-Gaussian models. *Precise setting:* With known linear dynamics, Gaussian initial state and Gaussian white process/measurement noise, Gaussian conditioning yields recursive conditional means and covariances.

Why it matters here: Estimate trajectories from noisy population observations.

Result 89: Takens' delay-embedding theorem. *Precise setting:* For a generic smooth observation of a compact d-dimensional manifold and generic diffeomorphism, sufficiently many delay coordinates (classically $2d + 1$) embed its state manifold.

Why it matters here: Understand the idealized mathematical conditions for reconstructing dynamics from scalar time series.

Result 90: Whitney embedding theorem. *Precise setting:* Every smooth d-dimensional manifold embeds smoothly into $\mathbb{R}^{2d}$ (with refinements and specific hypotheses).

Why it matters here: Motivate low-dimensional state representations, not neuronal sampling guarantees.

Result 91: Implicit function theorem. *Precise setting:* If a differentiable constraint $F(x, y) = 0$ has an invertible derivative in y at a point, it defines y locally as a differentiable function of x.

Why it matters here: Understand local tangent structure of constraint-defined manifolds.

Result 92: Tikhonov regularization convergence results. *Precise setting:* For a bounded linear inverse problem in Hilbert spaces, with proper regularization conditions and parameter choice relative to noise, regularized estimates converge to an appropriate minimum-norm solution as noise vanishes.

Why it matters here: Describe what stable lead-field inversion can recover and when source nonuniqueness remains.

Definition 17.1. A linear state-space model has

$$x_{t+1} = Ax_t + Bu_t + w_t, \qquad y_t = Cx_t + v_t.$$

The dimensions, covariance models, initial distribution, and observation mechanism must be declared before inferential conclusions.

Interpretation warning 17.2. Whitney and Takens do not guarantee that an arbitrary sparse neuron sample recovers the whole circuit manifold. Observability, measurement design, noise, and sampling density are independent concerns. Similarly, a regularized LFP inverse is not a unique physiological current reconstruction without additional constraints.

17.1 Proofs of selected dynamical results

Proof of Result 82 (Gronwall, integral form). Let u be continuous and nonnegative with $u(t) \le a + \int_0^t b\,u(s)\,ds$ for constants $a, b \ge 0$. Set $U(t) = a + \int_0^t b\,u(s)\,ds$. Then $U' = bu \le bU$, so $(e^{-bt}U)' \le 0$ and $U(t) \le ae^{bt}$. Hence $u(t) \le U(t) \le ae^{bt}$. □

Proof of Result 81 (Picard–Lindelöf), global Lipschitz case. Suppose $|f(t,x)-f(t,y)| \le L|x-y|$. On $C([0,h])$ with the sup norm, define $(\Phi x)(t) = x_0 + \int_0^t f(s,x(s))\,ds$. Then $\|\Phi x - \Phi y\|_\infty \le Lh\|x-y\|_\infty$, a contraction for $h < 1/L$. By Banach's theorem (Result 24) there is a unique fixed point, which is the unique solution on $[0,h]$. Repeating from $t = h$ extends it. Uniqueness on the whole interval also follows from Gronwall: two solutions satisfy $|x(t)-y(t)| \le \int_0^t L|x-y|\,ds$, so $a = 0$ forces $x = y$. □

Gronwall also quantifies sensitivity: if two initial conditions differ by δ, the trajectories differ by at most δe^{Lt}. For a chaotic network this bound is nearly attained, and predictions of individual trajectories lose accuracy exponentially, though statistical properties may still be predictable.

17.2 Lyapunov functions for linear systems

Theorem 17.3. *For $\dot{x} = Ax$, the origin is asymptotically stable iff for some (and then every) $Q \succ 0$ the Lyapunov equation $A^\top P + PA = -Q$ has a solution $P \succ 0$.*

Proof of sufficiency. Take $V(x) = x^\top P x > 0$ for $x \neq 0$. Along trajectories, $\dot{V} = x^\top(A^\top P + PA)x = -x^\top Q x < 0$ for $x \neq 0$. With $\lambda_{\min}(Q)\|x\|^2 \le x^\top Q x$ and $V \le \lambda_{\max}(P)\|x\|^2$, $\dot{V} \le -cV$ with $c = \lambda_{\min}(Q)/\lambda_{\max}(P)$, so $V(t) \le V(0)e^{-ct} \to 0$, which is Result 83 in quantitative form. □

For stable A, the solution is $P = \int_0^\infty e^{A^\top t} Q e^{At}\,dt$. The same equation with noise appears in stochastic models: for $dx = Ax\,dt + B\,dW$, the stationary covariance Σ solves $A\Sigma + \Sigma A^\top + BB^\top = 0$, which links the dynamics matrix fitted to neural data to the covariance of its fluctuations.

17.3 Observability and controllability proofs

Proof of Result 86. In discrete time, the noise-free outputs from x_0 are $y_k = CA^k x_0$. Stacking $k = 0, \dots, n-1$ gives $\mathcal{O}x_0$ with the observability matrix $\mathcal{O}$. If $\operatorname{rank}\mathcal{O} = n$, x_0 is determined. Conversely, if $\mathcal{O}x_0 = 0$ for some $x_0 \neq 0$, the Cayley–Hamilton theorem expresses every A^k, $k \ge n$, as a combination of $I, \dots, A^{n-1}$, so $CA^k x_0 = 0$ for all k: the state x_0 is invisible forever. □

Controllability (Result 87) is the dual statement for the pair $(A^\top, B^\top)$. In a recorded population, a latent direction outside the observable subspace cannot be estimated from that population no matter how long one records; in a stimulation experiment, a direction outside the controllable subspace cannot be driven by that stimulus.

17.4 Delay embedding in practice

Takens' theorem (Result 89) guarantees that the map $x_t \mapsto (y_t, y_{t-\tau}, \dots, y_{t-(m-1)\tau})$ is an embedding for generic smooth observations when $m \ge 2d + 1$. Practice adds three obstacles absent from the theorem: noise, which blurs the reconstructed manifold at small scales; finite data, which may not cover the attractor; and the choice of τ, for which common heuristics are the first zero of the autocorrelation or the first minimum of the mutual information. Methods

that infer causality from embeddings (convergent cross mapping) assume deterministic dynamics and fail with strong stochastic forcing or synchrony between the variables.

Exercises for this chapter

Exercise 17.4. Show that $\dot{x} = x^{2/3}$, $x(0) = 0$, has two solutions and identify which hypothesis of Result 81 fails.

Exercise 17.5. Solve $A^{\top}P + PA = -I$ for $A = \operatorname{diag}(-1, -2)$ and verify $P \succ 0$.

Exercise 17.6. For $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, determine which of $C = (1, 0)$ and $C = (0, 1)$ makes the pair observable.

Exercise 17.7. Compute the stationary covariance of $dx = -x\,dt + dW$ from the Lyapunov equation and compare with the Ornstein–Uhlenbeck formula.

Chapter 18

Oscillators, bifurcations, and stochastic dynamics

Oscillatory-looking trajectories can emerge from a dynamical system, an externally driven filter, or chance structure in a finite recording. These are mathematically distinct accounts.

18.1 Hopf normal form and an actual periodic attractor

The complex-valued ordinary differential equation $\dot{z} = (\mu + i\omega)z - |z|^2 z$ with $\omega \neq 0$ can be written $z = re^{i\theta}$, giving

$$\dot{r} = \mu r - r^3, \qquad \dot{\theta} = \omega.$$

For $\mu < 0$ the origin attracts nearby trajectories. For $\mu > 0$ the origin repels, while the circle $r = \sqrt{\mu}$ is an attracting periodic orbit. Its period is $2\pi/|\omega|$. This explicit derivation demonstrates a supercritical Hopf-type normal form; deciding whether experimental LFP oscillations arise from an actual Hopf bifurcation requires more evidence.

18.2 Phase reduction and weak coupling

A stable limit-cycle oscillator with phase coordinate θ responds to weak perturbation $u(t)$ to leading order as

$$\dot{\theta} = \omega + Z(\theta) \cdot u(t),$$

where Z is the infinitesimal phase-response curve (PRC). For two weakly coupled oscillators, averaging can produce a phase-difference equation $\dot{\psi} = \Delta\omega + \varepsilon H(\psi)$. Locking requires a stable fixed point of this equation; oscillatory phase locking need not imply anatomical synapses, and non-sinusoidal harmonics can generate apparent PAC.

18.3 jPCA and skew-symmetric dynamics

If $\dot{x} = Mx$ for a real skew-symmetric matrix $M^\top = -M$, then

$$\frac{d}{dt}\|x\|_2^2 = 2x^\top Mx = 0.$$

A real skew-symmetric matrix has orthogonal real normal form consisting of planar blocks $\left(\begin{smallmatrix} 0 & -\omega_j \\ \omega_j & 0 \end{smallmatrix}\right)$ (and possibly zeros); its nonzero eigenvalues occur as imaginary pairs $\pm i\omega_j$. Thus a fitted rotational component conserves Euclidean energy in the ideal autonomous model. The file `jpca_dynamics.m` fits such a skew generator to finite-difference derivatives [7], not a complete state-space model or a proof of physiological cycles.

18.4 Ornstein–Uhlenbeck noise and the Fokker–Planck equation

For an Itô stochastic differential equation $dX_t = a(X_t)dt + b(X_t)dW_t$, a twice differentiable function f obeys Itô's formula

$$df(X_t) = \left(af' + \tfrac{1}{2}b^2 f''\right)dt + bf'\,dW_t.$$

Its density, under regularity and boundary assumptions, evolves by the Fokker–Planck equation $\partial_t p = -\partial_x(ap) + \frac{1}{2}\partial_{xx}(b^2 p)$. For the Ornstein–Uhlenbeck process $dX_t = -\kappa X_t dt + \sigma dW_t$, with $\kappa > 0$, the stationary variance is $\sigma^2/(2\kappa)$ and the stationary autocovariance is $\sigma^2 e^{-\kappa|\tau|}/(2\kappa)$. An exponential autocorrelation is therefore compatible with a stochastic relaxation process and need not imply an intrinsic neuronal pacemaker or periodic oscillator.

Example 18.1 (Simulation check)**.** Simulate a damped OU fluctuation and a noisy Hopf limit cycle at the same sampling frequency. Compare autocorrelation, PSD and phase stability under different recording lengths. A narrow spectral peak alone does not establish the Hopf model.

18.5 Locking to a periodic input: the Adler equation

Let an oscillator with natural frequency ω be driven weakly at frequency Ω with a sinusoidal phase interaction. Averaging gives for the phase difference $\psi = \theta - \Omega t$

$$\dot{\psi} = \Delta\omega - \varepsilon \sin\psi, \qquad \Delta\omega = \omega - \Omega.$$

Fixed points exist iff $|\Delta\omega| \le \varepsilon$, at $\sin\psi^* = \Delta\omega/\varepsilon$; the one with $\cos\psi^* > 0$ is stable. Inside this range the oscillator is *entrained*: it runs at the driving frequency with a constant phase lag that depends on the detuning. Outside it, ψ drifts, with mean drift frequency $\sqrt{\Delta\omega^2 - \varepsilon^2}$ (phase slips separated by long quasi-locked epochs). Plotted in the (Ω, ε) plane, the locking region is a wedge, the first *Arnold tongue*; higher-order tongues for $p{:}q$ locking appear with nonsinusoidal interaction functions.

The equation predicts what to look for in rhythmic stimulation experiments: entrainment should hold over a range of stimulation frequencies that grows with intensity, the phase lag should vary systematically across that range, and the effect should vanish abruptly outside it. A response that simply follows every stimulus frequency with constant lag is more consistent with a driven filter than with an entrained oscillator.

18.6 The Kuramoto model

For N oscillators with natural frequencies ω_j drawn from a density g and all-to-all coupling,

$$\dot{\theta}_j = \omega_j + \frac{K}{N}\sum_{k=1}^{N} \sin(\theta_k - \theta_j).$$

With the order parameter $re^{i\Phi} = \frac{1}{N}\sum_k e^{i\theta_k}$ the system becomes $\dot\theta_j = \omega_j + Kr\sin(\Phi - \theta_j)$: each oscillator is driven by the mean field, an Adler equation with $\varepsilon = Kr$. Oscillators with $|\omega_j - \Omega| \le Kr$ lock; the others drift. Self-consistency of r gives Kuramoto's result: for a symmetric unimodal g, the incoherent state $r = 0$ loses stability at

$$K_c = \frac{2}{\pi g(0)},$$

and for a Lorentzian g of half-width γ, $K_c = 2\gamma$ and $r = \sqrt{1 - K_c/K}$ for $K > K_c$. The order parameter r is the population analogue of the phase-locking value, and the model shows that partial synchrony, a fraction of locked units with others drifting, is a generic state rather than a sign of measurement failure.

18.7 Center manifolds and the Hopf normal form

The normal form of this chapter is not an arbitrary model: near a bifurcation, it is what any smooth system reduces to.

Theorem 18.2 (Center manifold, stated)**.** *Let $\dot x = f(x,\mu)$ with $f(0,\mu) = 0$, and suppose $Df(0,0)$ has a pair of eigenvalues $\pm i\omega_0$ and all others with negative real part. Then for μ near 0 there is a locally invariant two-dimensional manifold through the origin that attracts nearby trajectories, and the dynamics on it are smoothly equivalent to $\dot z = (\alpha(\mu) + i\omega(\mu))z + \ell_1|z|^2 z + O(|z|^4)$.*

The sign of the real part of the *first Lyapunov coefficient* ℓ_1 decides the type: negative gives the supercritical Hopf of this chapter (small limit cycles growing as $\sqrt{\mu}$); positive gives a subcritical Hopf, in which a large-amplitude oscillation appears abruptly and coexists with the resting state (bistability, hysteresis). Neuron models distinguish these: class-2 excitability with a subcritical Hopf begins firing at a nonzero frequency and can show bistability between rest and repetitive firing.

18.8 Noise-driven oscillations near a Hopf point

Below the bifurcation ($\mu < 0$), noise $dz = (\mu + i\omega)z\,dt + \sigma\,dW$ (complex W) produces a stationary Gaussian process whose spectrum is a Lorentzian peaked at $\omega/(2\pi)$ with half-width $|\mu|/(2\pi)$ in hertz. Its amplitude has a Rayleigh distribution and its phase diffuses. Above the bifurcation, the amplitude concentrates near $\sqrt{\mu}$ and the phase still diffuses, with a much narrower peak. The two regimes can produce similar power spectra; the amplitude distribution and the duration of phase coherence discriminate them better than the spectrum.

Exercises for this chapter

Exercise 18.3. Solve the Adler equation outside the locking range and derive the mean drift frequency $\sqrt{\Delta\omega^2 - \varepsilon^2}$.

Exercise 18.4. For a Lorentzian g with $\gamma = 1\,\mathrm{rad/s}$, compute K_c and the order parameter at $K = 4$.

Exercise 18.5. Show that the OU process $dX = -\kappa X\,dt + \sigma\,dW$ has spectral density proportional to $1/(\kappa^2 + (2\pi f)^2)$.

Exercise 18.6. For $\dot{r} = \mu r + r^3 - r^5$, find the equilibria and their stability, and describe the hysteresis as μ crosses zero.

Chapter 19

Robust estimates, information bias, and numerical stability

Analysis pipelines need both statistical robustness to unusual data points and numerical robustness to finite precision. Neither follows merely from a mathematical formula being valid.

19.1 Median and median absolute deviation

The sample median has finite-sample replacement breakdown point approaching 1/2: fewer than about half the observations cannot drive it arbitrarily far by contamination. For observations $x_1, \ldots, x_T$, define MAD $=$ median$_i\,|x_i - \text{median}_j\, x_j|$. For normal errors a common robust scale estimate is $1.4826\,\text{MAD}$, asymptotically calibrated under the Gaussian model. For artifacted spike waveforms, median/MAD thresholding can be more stable than a mean/standard deviation, but physiological artifacts and nonstationarity can contaminate even robust estimates. An influence function formalizes the estimator's infinitesimal response to a point-mass contamination.

19.2 Empirical distribution bounds and time-rescaling

For iid observations with distribution function F, let F_T be the empirical CDF. A Dvoretzky–Kiefer–Wolfowitz bound states for $\epsilon > 0$

$$\mathbb{P}\left(\sup_x |F_T(x) - F(x)| > \epsilon\right) \leq 2e^{-2T\epsilon^2}.$$

It implies uniform consistency (a Glivenko–Cantelli conclusion) in the iid setting. The KS check of rescaled interspike intervals uses an exponential or uniform null only when the conditional intensity is correctly specified and relevant transformed intervals satisfy the needed independence assumptions. Repeated fitting and testing on the same data changes calibration.

19.3 Information-theoretic chain rules and finite-sample bias

For discrete variables, the entropy chain rule is $H(X,Y) = H(X) + H(Y \mid X)$ and $I(X;Y \mid Z) = H(X \mid Z) - H(X \mid Y,Z)$. The Gaussian distribution maximizes differential entropy among random vectors with a specified positive-definite covariance, with entropy $\frac{1}{2}\log[(2\pi e)^d \det \Sigma]$. A plug-in entropy estimator from a finite discrete histogram usually has downward bias. With K effectively occupied bins, a leading Miller–Madow-style correction in *nats* is $(K-1)/(2T)$ under its asymptotic sampling assumptions. It does not cure curse-of-dimensionality or guarantee an unbiased information estimator. Continuous estimators such as KSG have different nearest-neighbor assumptions and biases; the `mutual_information.m` implementation should not be described as using KSG without verifying code.

19.4 Conditioning, least squares, and stable algorithms

If a full-rank matrix X has spectral condition number $\kappa_2(X) = \sigma_{\max}/\sigma_{\min}$, then $\kappa_2(X^\top X) = \kappa_2(X)^2$.

Proof. The singular values of X are the square roots of the eigenvalues of $X^\top X$; hence those eigenvalues are σ_j^2 and the ratio of extremes is squared. □

This explains why explicitly solving normal equations can amplify roundoff relative to QR/SVD solutions. A backward-stable algorithm returns an exact answer for a nearby problem; it does not guarantee small forward error for ill-conditioned inputs. The FFT reduces the computation of a DFT from the naive $O(T^2)$ complex arithmetic to $O(T \log T)$ for appropriate factorizable sizes. Algorithmic efficiency does not change spectral-estimator uncertainty.

Example 19.1 (Research reproducibility)**.** When regressing spike history on population counts, report condition numbers, convergence criteria, effective independent units, model selection, bin duration and handling of omitted variables, not just a coefficient array.

19.5 Influence functions and M-estimators

For an estimator defined as a functional $T(F)$ of the distribution, the *influence function* at x is $\mathrm{IF}(x) = \lim_{\epsilon \downarrow 0}[T((1-\epsilon)F + \epsilon\delta_x) - T(F)]/\epsilon$. For the mean, $\mathrm{IF}(x) = x - \mu$: unbounded, so a single artefact can move it arbitrarily. For the median of a continuous F with density p, $\mathrm{IF}(x) = \mathrm{sign}(x-m)/(2p(m))$: bounded. Huber's M-estimator of location minimizes $\sum_i \rho_c(x_i - \theta)$ with

$$\rho_c(u) = \begin{cases} u^2/2, & |u| \le c, \\ c|u| - c^2/2, & |u| > c, \end{cases}$$

which behaves like the mean for small residuals and like the median for large ones; its influence function is $\psi_c(u) = \max(-c, \min(c,u))$ up to scaling. With $c = 1.345\sigma$, it retains 95% efficiency at the Gaussian while bounding the influence of artefacts.

19.6 Derivation of the Miller–Madow correction

For a discrete distribution $p_1, \dots, p_K$ and empirical frequencies $\hat{p}_i$ from T samples, expand the plug-in entropy $\hat{H} = -\sum_i \hat{p}_i \log \hat{p}_i$ to second order around p. The first-order term has zero mean, and the second-order term is $-\sum_i (\hat{p}_i - p_i)^2/(2p_i)$. Since $\mathbb{E}(\hat{p}_i - p_i)^2 = p_i(1-p_i)/T$,

$$\mathbb{E}\hat{H} \approx H - \sum_i \frac{1 - p_i}{2T} = H - \frac{K-1}{2T}.$$

The bias is negative, grows with the number of occupied bins, and shrinks only as $1/T$. Mutual information estimated as $\hat{H}(X) + \hat{H}(Y) - \hat{H}(X,Y)$ inherits a *positive* bias of about $(K_X - 1)(K_Y - 1)/(2T)$ nats, because the joint histogram has many more bins. With 10 response bins, 8 stimuli and 200 trials, the bias is about $63/400 \approx 0.16$ nats, comparable to many reported effects. Shuffle correction (subtracting the information computed after permuting stimulus labels) estimates and removes this bias empirically.

19.7 Confidence bands from DKW

Setting the DKW bound equal to α gives a simultaneous confidence band for an entire distribution function:

$$\mathbb{P}\Big(\sup_x |F_T(x) - F(x)| \le \epsilon_T\Big) \ge 1 - \alpha, \qquad \epsilon_T = \sqrt{\frac{\log(2/\alpha)}{2T}}.$$

For $T = 400$ rescaled intervals and $\alpha = 0.05$, $\epsilon_T \approx 0.068$. This band is valid for every T, not only asymptotically, and here it has essentially the same half-width as the asymptotic KS band $1.36/\sqrt{T} = 0.068$. The DKW band applies to iid samples; for time-rescaled intervals it presupposes a correctly specified model, as stated above.

19.8 Stable evaluation of log-likelihoods

Likelihoods of long spike trains or of mixtures involve products of many small numbers and sums of exponentials. Three rules prevent overflow and underflow:

1. work with log-likelihoods, never likelihoods;
2. compute $\log \sum_i e^{a_i}$ as $a^* + \log \sum_i e^{a_i - a^*}$ with $a^* = \max_i a_i$ (the *log-sum-exp* trick), which is exact in exact arithmetic and never overflows;
3. compute $\log(1 + x)$ and $e^x - 1$ for small x with dedicated functions (`log1p`, `expm1`) to avoid cancellation, as in the complementary log-log link.

Example 19.2 (Posterior over 1000 hypotheses)**.** A Bayesian decoder evaluates the log-likelihood of a population response under 1000 candidate positions, with values around -5000. Direct exponentiation underflows to zero for all of them and gives $0/0$. Subtracting the maximum first gives finite weights, and the posterior is recovered exactly.

Exercises for this chapter

Exercise 19.3. Compute the breakdown point of the trimmed mean that discards the largest and smallest 10% of observations.

Exercise 19.4. Derive the second-order bias of the plug-in entropy estimator in detail.

Exercise 19.5. Find T such that the DKW band at $\alpha = 0.05$ has half-width 0.02.

Exercise 19.6. Show that the log-sum-exp identity holds for any choice of a^*, and explain why the maximum is the safe choice.

Part V

Connections and worked mathematics

Chapter 20

Worked proofs and derivations

We now move from cataloguing results to deriving selected statements. Every example connects a small mathematical argument with a possible analysis of electrophysiological recordings.

20.1 Proving a set identity

Fix arbitrary x. By definition of union, $x \notin A \cup B$ means that $x \notin A$ and $x \notin B$, which means $x \in A^c \cap B^c$. Thus $(A \cup B)^c = A^c \cap B^c$. Equality of sets was proved by equivalence of membership statements.

20.2 Distinguishing relations from functions

Let U be a set of identified units and T an observation interval. Spike occurrences form a relation $R \subseteq U \times T$. The relation is generally not a function $U \to T$ because one unit may have multiple spike times. It does define $F : U \to \mathcal{P}(T)$ by $F(u) = \{t \in T : (u, t) \in R\}$. The codomain $\mathcal{P}(T)$ is the set of subsets of T.

20.3 Cauchy–Schwarz from an orthogonal projection

For vectors x, y in a real inner-product space, $y \neq 0$,

$$0 \leq \left\| x - \frac{\langle x, y\rangle}{\|y\|^2} y \right\|^2 = \|x\|^2 - \frac{\langle x, y\rangle^2}{\|y\|^2}.$$

Rearrangement yields $|\langle x, y\rangle| \leq \|x\|\|y\|$. The complex case inserts conjugates as dictated by the inner-product convention. On centered square-integrable random variables, this yields $|\operatorname{Cov}(X, Y)|^2 \leq \operatorname{Var}(X) \operatorname{Var}(Y)$. Thus $|\rho_{XY}| \leq 1$ for positive variances, while a constant variable gives undefined rather than zero correlation.

20.4 Least squares as orthogonal projection

For $X \in \mathbb{R}^{T \times p}$ with full column rank, minimize $L(\beta) = \|y - X\beta\|_2^2$. Differentiating gives

$$\nabla L(\beta) = -2X^\top (y - X\beta), \qquad X^\top X \widehat{\beta} = X^\top y.$$

The fitted residual $r = y - X\widehat{\beta}$ is orthogonal to every column of X. One can write $\widehat{\beta} = (X^\top X)^{-1} X^\top y$ under the rank assumption; numerically QR/SVD or stable linear solvers are preferable to explicit inversion.

20.5 PCA as a constrained maximization

Center the rows of $X \in \mathbb{R}^{T \times N}$ with $T > 1$ and form $C = X^\top X/(T-1)$. The projected sample variance along unit vector v is $v^\top C v$. With a Lagrange multiplier λ,

$$\mathcal{L}(v, \lambda) = v^\top C v - \lambda(v^\top v - 1), \qquad Cv = \lambda v.$$

By the spectral theorem, maximizing the Rayleigh quotient selects a leading eigenvector; orthogonality constraints yield subsequent components. This proves PCA optimality relative to *sample variance*, not physiological importance.

20.6 Participation-ratio bounds

For r strictly positive covariance eigenvalues, Cauchy–Schwarz gives $(\sum_i \lambda_i)^2 \le r \sum_i \lambda_i^2$. Because all cross products $\lambda_i \lambda_j$ are nonnegative, $(\sum_i \lambda_i)^2 \ge \sum_i \lambda_i^2$. Therefore

$$1 \le \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} \le r.$$

Equality on the right holds when positive eigenvalues are equal; for $r > 1$, the lower bound is approached if one eigenvalue dominates and the remaining ones vanish relative to it.

20.7 Why a Poisson mixture is overdispersed

Let $G > 0$, $\mathbb{E}G = 1$, and $N \mid G \sim \text{Poisson}(\mu G)$. The conditional mean and variance both equal μG. Total expectation yields $\mathbb{E}N = \mu$; total variance gives

$$\text{Var}(N) = \mathbb{E}[\mu G] + \text{Var}(\mu G) = \mu + \mu^2 \,\text{Var}(G).$$

Thus $\text{Var}(N)/\mathbb{E}N = 1 + \mu \,\text{Var}(G)$. This is a model-based result, not proof that actual response variability comes from a particular neural gain mechanism [13].

20.8 Coherence and spectral positive semidefiniteness

Suppose the spectral matrix is

$$S(f) = \begin{pmatrix} S_{xx}(f) & S_{xy}(f) \\ S_{xy}(f)^* & S_{yy}(f) \end{pmatrix} \succeq 0.$$

The determinant must be nonnegative: $S_{xx}S_{yy} - |S_{xy}|^2 \ge 0$. If both auto-spectra are positive, $0 \le |S_{xy}|^2/(S_{xx}S_{yy}) \le 1$. This proves a normalized spectral dependence bound, not biophysical connectivity.

20.9 Conditional point-process likelihood

If $\lambda(t \mid \mathcal{H}_t)$ is a valid predictable conditional intensity and $t_1, \dots, t_n$ are observed events in $(0, T]$, the point-process log-likelihood relative to a specified reference measure is, up to reference terms,

$$\ell = \sum_{i=1}^{n} \log \lambda(t_i \mid \mathcal{H}_{t_i}) - \int_0^T \lambda(s \mid \mathcal{H}_s)\, ds.$$

The event term rewards model intensity at observed spikes; the compensator penalizes excessive predicted spikes. It follows by combining the small-bin event and no-event probabilities under regularity conditions.

20.10 Predictive influence is not direct physical causation

Let z_t be a latent driver and define observed series $x_t = z_t + \epsilon_t$ and $y_t = z_{t-1} + \eta_t$. Past x can help predict y, yet this construction contains no direct physical edge from x to y. Consequently Granger prediction depends on the chosen variables and may reflect unobserved common inputs [4, 6].

20.11 The time-rescaling theorem for a conditional intensity

Theorem 20.1. *Let N be a simple point process on $[0, T]$ with continuous, strictly positive conditional intensity $\lambda(t \mid \mathcal{H}_t)$, and let $\Lambda(t) = \int_0^t \lambda(s \mid \mathcal{H}_s)\, ds$. Then the rescaled times $\Lambda(t_1) < \Lambda(t_2) < \cdots$ form a unit-rate Poisson process on $[0, \Lambda(T)]$.*

Proof via the likelihood. By Result 76, the joint density of the first n event times is $\prod_{i=1}^{n} \lambda(t_i \mid \mathcal{H}_{t_i}) \exp(-\Lambda(t_n))$. Change variables to $z_i = \Lambda(t_i) - \Lambda(t_{i-1})$. Because Λ is differentiable with derivative λ, and because $\Lambda(t_i)$ depends only on events before t_i, the Jacobian matrix of the map $(t_1, \dots, t_n) \mapsto (z_1, \dots, z_n)$ is triangular with diagonal entries $\lambda(t_i \mid \mathcal{H}_{t_i})$. The density of $(z_1, \dots, z_n)$ is therefore $\exp(-\sum_i z_i)$ on the positive orthant: the z_i are iid exponential with mean one. □

The proof shows exactly where predictability is used: the triangular Jacobian requires that the intensity at t_i not depend on t_i itself or on later events. A model whose "history" term accidentally uses the current bin's spike passes every goodness-of-fit test trivially.

20.12 Gauss–Markov and whitening

If $y = X\beta + \varepsilon$ with $\operatorname{Cov} \varepsilon = \sigma^2 \Omega$, $\Omega \succ 0$ known, set $\tilde{y} = \Omega^{-1/2} y$ and $\tilde{X} = \Omega^{-1/2} X$. The transformed errors have covariance $\sigma^2 I$, so by Result 31 the best linear unbiased estimator is OLS on the transformed data:

$$\hat{\beta}_{\mathrm{GLS}} = (X^\top \Omega^{-1} X)^{-1} X^\top \Omega^{-1} y.$$

For AR(1) errors with coefficient a, $\Omega^{-1/2}$ is the prewhitening filter $\tilde{y}_t = y_t - a y_{t-1}$ (with a corrected first sample). Regression of a slowly varying LFP feature on behaviour with OLS gives unbiased but inefficient coefficients and badly overconfident standard errors; prewhitening fixes both if a is well estimated.

20.13 Positive semidefiniteness of empirical spectral matrices

Proposition 20.2. *For any data, the multitaper or Welch cross-spectral estimate* $\hat{S}(f) = \frac{1}{K}\sum_{k=1}^{K} z_k(f) z_k(f)^*$*, with* $z_k(f)$ *the vector of tapered Fourier coefficients across channels, is Hermitian positive semidefinite. Consequently the estimated coherence lies in* $[0, 1]$*.*

Proof. Each $z_k z_k^*$ is Hermitian with $a^* z_k z_k^* a = |z_k^* a|^2 \geq 0$; a nonnegative combination preserves both properties. Apply the 2×2 determinant argument used for Result 69. □

With $K = 1$ (a single taper and trial), $\hat{S}$ has rank one and the estimated coherence equals one at every frequency for any pair of signals. Coherence estimation requires averaging, and the number of averaged terms sets the bias floor $\approx 1/K$.

20.14 An identity behind trial-averaged evoked responses

Let $x_r(t) = s(t) + n_r(t)$ for trials $r = 1, \ldots, R$ with zero-mean noise of variance $\sigma^2(t)$, independent across trials. The trial average $\bar{x}$ has variance σ^2/R, and

$$\mathbb{E}\Big[\frac{1}{R-1}\sum_{r}(x_r - \bar{x})^2\Big] = \sigma^2, \qquad \mathbb{E}[\bar{x}^2] = s^2 + \frac{\sigma^2}{R}.$$

Hence the power of the average overestimates the power of the evoked signal by σ^2/R, and the unbiased estimate is $\bar{x}^2 - \hat{\sigma}^2/R$. Comparing evoked power between conditions with different trial counts without this correction biases the comparison towards the condition with fewer trials.

Chapter 21

Theorems and the existing MATLAB arsenal

The paths below are checked against the repository's `matlab/path/` directory at the start of this edition. Each source file is an exploratory primitive rather than a guarantee of complete replication of the published method. Consult `matlab/METHODS.tsv` for the current implementation statuses.

Mathematical prerequisite	MATLAB source (under matlab/path)	Analysis and caution
Sampling, convolution, Hilbert and Bedrosian	`bandpass_lfp.m`, `analytic_phase.m`	Signal phase depends on justified filtering; nonstationarity can create apparent PAC.
Wiener–Khinchin, Parseval	`welch_psd.m`, `coherence_spectrum.m`	Finite-window PSD/coherence estimates, not exact population spectra.
Slepian spectral concentration	`multitaper_psd.m`, `multitaper_coherence.m`	DPSS estimates; estimator design and inference require further checks.
Periodic / aperiodic modeling	`aperiodic_slope.m`, `irasa_decomposition.m`	Exploratory methods; fitting a background does not certify an oscillation.
Circular statistics	`circular_mean.m`, `rayleigh_test.m`	Circular inferential assumptions and dependence of phases matter.
Phase-amplitude and phase locking	`modulation_index.m`, `pac_mean_vector_length.m`, `phase_locking_value.m`	Non-sinusoidal harmonics and surrogates need explicit controls.
Randomization inference	`paired_signflip_test.m`, `jitter_spike_surrogate.m`	Exchangeability or an explicit conditional jitter null is required.

Mathematical prerequisite	MATLAB source (under matlab/path)	Analysis and caution
Spectral theorem and SVD	`population_pca.m`, `assembly_pca.m`	Variance-optimal directions need not identify biological sources.
Effective spectral dimension	`participation_ratio.m`, `eigenspectrum_powerlaw.m`	Participation ratio does not automatically equal manifold dimension.
Low-rank regression	`reduced_rank_regression.m`, `communication_subspace_projection.m`	Cross-validated prediction needed before mechanistic interpretation.
Tensor decomposition	`tensor_cp_als.m`	CP-ALS primitive is not a complete TCA analysis pipeline.
Conditional expectation, variance	`spike_count_fano.m`, `noise_correlation.m`	Count windows and latent gain change sampling interpretations.
Point-process likelihood	`poisson_spike_glm.m`, `spike_history_design.m`	Binned IRLS approximation; missing latent drive and overdispersion checks.
Poisson event statistics	`spike_bin_counts.m`, `peri_event_histogram.m`	Temporal interval conventions and boundaries affect event counts.
Autoregressive prediction, Wold	`granger_causality.m`	Temporal VAR score; not spectral Granger or direct causation.
Linear dynamics / skew symmetry	`jpca_dynamics.m`	Skew-symmetric fit on prepared trajectories; not full published jPCA.
Linear inverse problems	`field_potential_inverse_tikhonov.m`	Regularized estimate, not uniquely identified neural currents.
Mutual information and DPI	`mutual_information.m`, `transfer_entropy.m`	Finite-sample bias, history embedding and discretization matter.

21.1 Theorems without a dedicated implementation

The catalogue also includes Kolmogorov extension, time-rescaling goodness-of-fit, Davis–Kahan perturbation, Takens embedding, and Kalman observability. Gaussian KL divergence and Gaussian conditioning are now derived explicitly in the Gaussian toolkit These are *mathematical prerequisites or future checks*, not claims that their complete scientific imple-

mentations exist in the repository. The presence of `gaussian_process_smoothing.m` must not be interpreted as a full GPFA model.

21.2 Operational contract

Read input shapes and units from the MATLAB function help. Document sample rate, time stamps, channel orientation, boundary conventions, electrode reference, trial alignment, and the meaning of each null hypothesis. Benchmark a method against simulated data of known ground truth before trying to infer physiology from an empirical recording.

21.3 From biological generators to filters and estimates

The physical forward chain has distinct operations: a cable/membrane source model supplies transmembrane currents; a conductivity model with boundary conditions constructs a lead field; `field_potential_forward.m` applies a *supplied* lead field; `bandpass_lfp.m` then uses Butterworth sections and forward–backward noncausal filtering; spectral estimators operate on that processed trace. Inverting a supplied lead field is regularized but not necessarily unique. No software primitive converts a fitted spectral peak directly into a uniquely identified membrane current.

21.4 Model-fitting limitations exposed by the new chapters

`poisson_spike_glm.m` uses a binned canonical Poisson model with clipped linear predictors: deriving the ideal log-concave Poisson objective does not guarantee that every numerical iterate of the implementation maximizes the same unclipped likelihood. `tensor_cp_als.m` is a ridge-regularized CP-ALS primitive; alternating steps are not a global-convergence theorem. `granger_causality.m` returns a temporal predictive log-variance ratio, while `transfer_entropy.m` uses discretized first-order conditional mutual information. The Gaussian equivalence theorem does not claim these distinct implementations have matching finite-sample outputs or identify causal interventions.

Chapter 22

Exercises, counterexamples, and research projects

The following problems are original exercises for the CON^{2}PHYS curriculum. The objective is to prove each mathematical claim first and only then write MATLAB experiments.

Exercise 22.1 (Relations and functions)**.** Define $R \subseteq U \times T$ for two units firing at the same timestamp. Prove that R need not be a function $U \to T$. Construct the corresponding function into $\mathcal{P}(T)$.

Exercise 22.2 (Sampling failure)**.** Sample a 90-Hz sine wave at 100 Hz and identify a lower-frequency sinusoid with the same samples. Explain why a digital filter applied after sampling cannot recover the true generating frequency.

Exercise 22.3 (Hilbert envelope)**.** Construct $x(t) = (1 + 0.3\cos 2\pi 2t)\cos 2\pi 40t$. Calculate the analytic envelope and compare against a signal whose envelope frequencies overlap its carrier spectrum. Explain when the Bedrosian assumption fails.

Exercise 22.4 (Spectral concentration)**.** Generate equal-length noisy signals with an off-bin sinusoid. Compare ordinary periodograms, Welch, and DPSS multitaper estimates using repeated simulations. Vary time-bandwidth product and report bias versus variance.

Exercise 22.5 (Principal subspace perturbations)**.** Generate a diagonal covariance with a controllable eigengap. Perturb it by symmetric noise, compute leading subspace angles, and compare their variation with the Davis–Kahan scaling.

Exercise 22.6 (Random-matrix nulls)**.** Simulate isotropic Gaussian matrices with independent samples, then introduce temporal autocorrelation and heterogeneous channel noise. Compare each empirical eigenvalue histogram with the corresponding idealized Marcenko–Pastur curve.

Exercise 22.7 (Latent gain)**.** Generate counts with $N \mid G \sim \mathrm{Poisson}(\mu G)$, where G is a positive random variable of mean one. Verify the predicted quadratic mean–variance relation and test how a purely Poisson GLM misestimates uncertainty.

Exercise 22.8 (Conditional point-process validation)**.** Fit a simulated point-process intensity with and without spike-history effects. Examine rescaled inter-event intervals. Explain why correctly transformed residuals do not identify a unique mechanism.

Exercise 22.9 (Shared-input Granger counterexample)**.** Simulate an unobserved driver of two time series with different lags. Show that adding measurements of the common source can

change inferred predictive direction or strength. Avoid interpreting this as proof of synaptic connectivity.

Exercise 22.10 (Inverse problem)**.** Construct a lead-field matrix with a nontrivial nullspace. Find two distinct current patterns with identical electrode predictions, then compare Tikhonov solutions for different regularization strengths.

22.1 Suggested cumulative projects

Project I: from set membership to a reliable spike-count representation. Project II: from convolution to LFP spectral density and multitaper estimators. Project III: from covariance to PCA stability and manifold caveats. Project IV: from conditional probability to spike-train GLM validation. Project V: from state-space observability to limits of causal and mechanistic inference.

22.2 Further problems by chapter

Exercise 22.11 (Partitions and sorting)**.** Two spike-sorting runs on the same data give partitions $\mathcal{P}$ and $\mathcal{Q}$. Prove that the number of spike pairs placed together by both runs equals $\sum_{P,Q} \binom{|P\cap Q|}{2}$, and use this to define the Rand index.

Exercise 22.12 (Contraction and recurrent rates)**.** For $r = \tanh(wr + I)$, find the range of w for which the fixed-point iteration is guaranteed to converge for every I.

Exercise 22.13 (PCA of a random walk)**.** Simulate 100 independent random walks of length 1000, compute the PCA across time, and compare the leading components with the Karhunen–Loève eigenfunctions of Brownian motion.

Exercise 22.14 (Delta method for coherence)**.** Using the Fisher transformation $\operatorname{atanh}|\hat{c}|$, derive an approximate confidence interval for coherence from K tapers and compare it with a jackknife over tapers in simulation.

Exercise 22.15 (GHK inversion)**.** Given reversal potentials measured under two different extracellular sodium concentrations, show that the ratio $P_{\mathrm{Na}}/P_{\mathrm{K}}$ is identifiable from the GHK voltage equation, while the absolute permeabilities are not.

Exercise 22.16 (Causal phase estimation)**.** Fit an AR model to the past 500 ms of an LFP and forecast 100 ms ahead to estimate the current theta phase. Compare the error with that of a causal Butterworth filter and with the zero-phase reference.

Exercise 22.17 (Hawkes overdispersion)**.** Simulate a Hawkes process with exponential kernel and branching ratio 0.5, and verify the long-window Fano factor $1/(1-\nu)^2$.

Exercise 22.18 (Collider bias in event-triggered analysis)**.** Simulate two independent Poisson neurons, define population events as bins where the summed count exceeds a threshold, and compute their correlation within and outside events.

Exercise 22.19 (Spiked covariance)**.** Plant a rank-one component of variance $1 + \theta$ in $N = 100$ isotropic Gaussian neurons. For $T \in \{50, 100, 400, 1600\}$, estimate the leading sample eigenvalue and the squared alignment, and compare with the BBP predictions.

Exercise 22.20 (Entrainment tongue)**.** Drive a noisy phase oscillator with natural frequency 8 Hz at frequencies from 6 to 10 Hz and several intensities. Map the region in which the phase-locking value to the drive exceeds 0.8, and compare with the Adler prediction.

Exercise 22.21 (Entropy bias)**.** For a uniform distribution on $K = 16$ symbols, estimate the plug-in entropy bias for $T = 20, 50, 200$ by simulation and compare with $(K - 1)/(2T)$.

Exercise 22.22 (Time rescaling with a wrong model)**.** Simulate an inhomogeneous Poisson process with sinusoidal rate, fit a constant-rate model, and show how the KS plot of rescaled intervals reveals the misspecification.

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